

Fourier series: Ex: 22, 23, 24, 25, 31
Problem-19

Line spectrum: Ex: 32, 33, 34

Convolution sum, convolution integral: Follow lecture sheet

Laplace Transform: 58, 60, 61, 64, 68, 72, 75,

Inverse Laplace Transform: Ex-101

Fourier transform: Ex-39, 40, 41, 43

Unit step function: ex-89, 91, 92

Ramp function: 90

Impulse differential equation using Laplace transform:

Ex-103, 104, 105

MATLAB

Example-22

$$y = f(t) = 0 ; -4 \leq t \leq 0$$

$$y = 5 ; 0 \leq t \leq 4$$

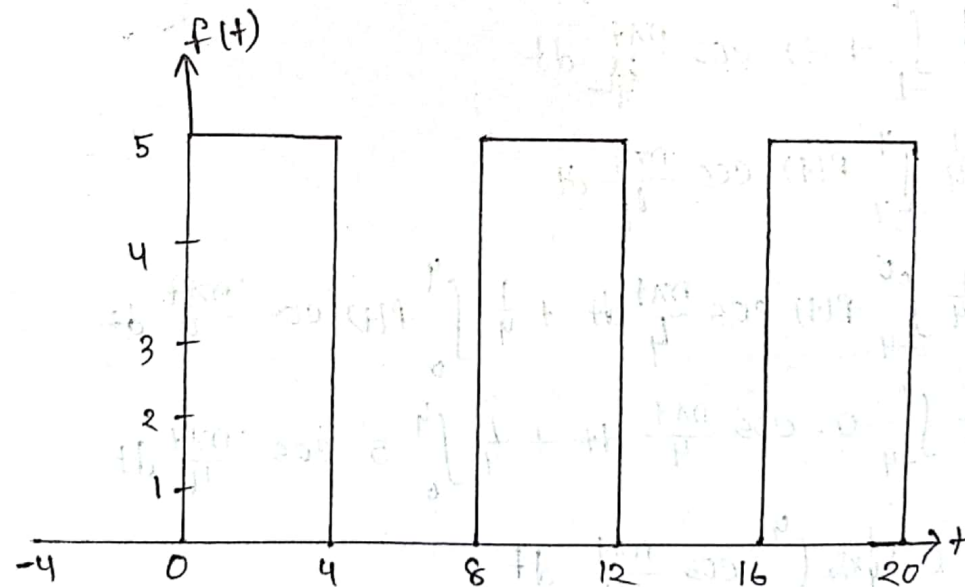
$$f(t) = f(t+8), \text{ Here } T=2L=8 \therefore L=4$$

(a) Sketch the function for 3 cycle.

(b) Find the Fourier series for the function.

Solution:

(a)



A periodic signal with period $T=2L=8$

(b)

$$a_0 = \frac{1}{L} \int_{-L}^L f(t) dt$$

$$= \frac{1}{4} \int_{-4}^4 f(t) dt$$

$$= \frac{1}{4} \int_{-4}^0 f(t) dt + \frac{1}{4} \int_0^4 f(t) dt$$

$$= \frac{1}{4} \int_{-4}^0 0 \cdot dt + \frac{1}{4} \int_0^4 5 dt \quad \left[\begin{array}{l} y = f(t) = 0 ; -4 \leq t \leq 0 \\ y = 5 ; 0 \leq t \leq 4 \end{array} \right]$$

$$= 0 + \frac{1}{4} [5t]_0^4$$

$$= \frac{1}{4} \times [5 \times 4 - 0]$$

$$= \frac{20}{4}$$

$$= 5$$

$$a_n = \frac{1}{L} \int_{-L}^L f(t) \cos \frac{n\pi t}{L} dt$$

$$= \frac{1}{4} \int_{-4}^4 f(t) \cos \frac{n\pi t}{4} dt$$

$$= \frac{1}{4} \int_{-4}^0 f(t) \cos \frac{n\pi t}{4} dt + \frac{1}{4} \int_0^4 f(t) \cos \frac{n\pi t}{4} dt$$

$$= \frac{1}{4} \int_{-4}^0 0 \cdot \cos \frac{n\pi t}{4} dt + \frac{1}{4} \int_0^4 5 \cos \frac{n\pi t}{4} dt$$

$$= 0 + \frac{1}{4} \times 5 \int_0^4 \cos \frac{n\pi t}{4} dt$$

$$= \frac{5}{4} \int_0^4 \cos \frac{n\pi t}{4} dt \quad \left[\int \cos mx \, dx = \frac{1}{m} \sin mx \right]$$

$$= \frac{5}{4} \times \frac{4}{n\pi} \left[\sin \frac{n\pi \times 4}{4} - \sin \frac{n\pi \times 0}{4} \right]$$

$$= \frac{5}{n\pi} [\sin n\pi - \sin 0]$$

$$= \frac{5}{n\pi} \sin n\pi$$

$$= 0 \quad [\because \sin \pi = \sin 2\pi = \sin 3\pi = \dots = \sin n\pi = 0 \text{ and } \sin 0 = 0]$$

$$\therefore b_n = \frac{1}{L} \int_{-L}^L f(t) \sin \frac{n\pi t}{L} dt$$

$$= \frac{1}{4} \int_{-4}^4 f(t) \sin \frac{n\pi t}{4} dt$$

$$= \frac{1}{4} \int_{-4}^0 f(t) \sin \frac{n\pi t}{4} dt + \frac{1}{4} \int_0^4 f(t) \sin \frac{n\pi t}{4} dt$$

$$= \frac{1}{4} \int_{-4}^0 (0) \sin \frac{n\pi t}{4} dt + \frac{1}{4} \int_0^4 (5) \sin \frac{n\pi t}{4} dt$$

$$= 0 + \frac{1}{4} \times 5 \int_0^4 \sin \frac{n\pi t}{4} dt$$

$$= \frac{1}{4} \times 5 \int_0^4 \sin \frac{n\pi t}{4} dt$$

$$= \frac{5}{4} \int_0^4 \sin \frac{n\pi t}{4} dt$$

$$= \frac{5}{4} \times \frac{1}{\frac{n\pi}{4}} \left[-\cos \frac{n\pi t}{4} \right]_0^4$$

$$= -\frac{5}{4} \times \frac{4}{n\pi} \left[\cos \frac{n\pi t}{4} \right]_0^4$$

$$= -\frac{5}{n\pi} \left[\cos \frac{n\pi \times 4}{4} - \cos \frac{n\pi \times 0}{4} \right]$$

$$= -\frac{5}{n\pi} [\cos n\pi - \cos 0]$$

$$= -\frac{5}{n\pi} [\cos n\pi - 1]$$

$$\left[\int \cos mx \, dx = \frac{1}{m} \sin mx; \right. \\ \left. \int \sin mx \, dx = -\frac{1}{m} \cos mx \right]$$

The fourier series for the above function:

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi t}{L} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi t}{L}$$

$$= \frac{5}{2} + \sum_{n=1}^{\infty} (0) \cos \frac{n\pi t}{4} + \sum_{n=1}^{\infty} \left[-\frac{5}{n\pi} (\cos n\pi - 1) \right] \sin \frac{n\pi t}{4}$$

$$= 2.5 - \frac{5}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} (\cos n\pi - 1) \sin \frac{n\pi t}{4}$$

$$\therefore f(t) = \underbrace{2.5}_{\text{DC Value}} - \underbrace{\left[\frac{5}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} (\cos n\pi - 1) \sin \frac{n\pi t}{4} \right]}_{\text{AC value}}$$

Complex wave

(Ans)

Example-29

$$y = f(t) = -1; \quad -\pi < t < 0$$

$$= 1; \quad 0 < t < \pi$$

$$f(t) = f(t + 2\pi) \quad \text{Hence, } T = 2L = 2\pi$$

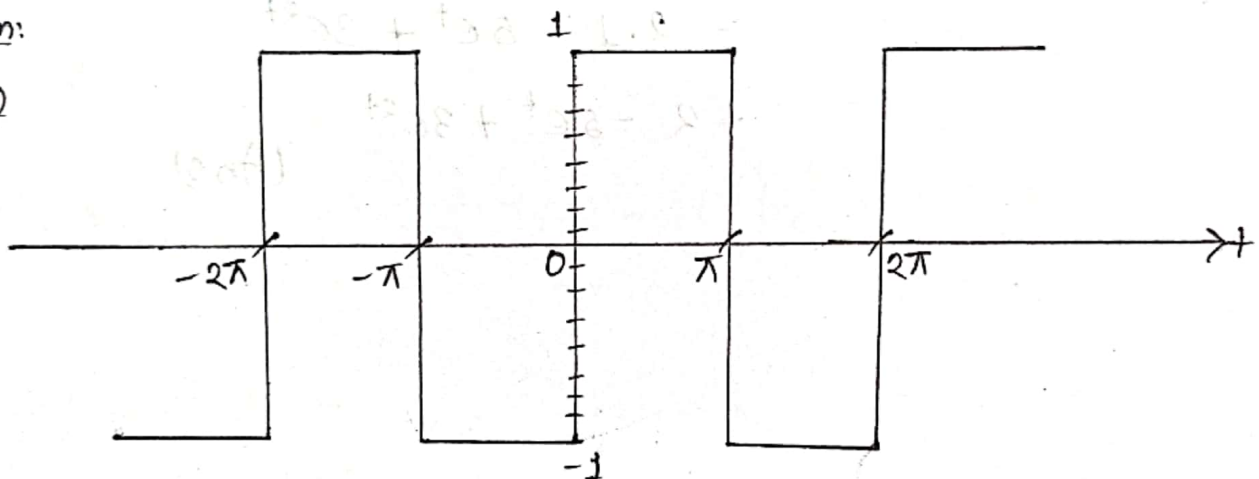
$$\therefore L = \pi$$

(a) sketch the function for 3 cycles:

(b) Find the fourier series for the function.

Solution:

(a)



A periodic signal with period $T = 2L = 2\pi$

(b) we know that,

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi t}{L} + b_n \sin \frac{n\pi t}{L} \right)$$

Given that,

$$y = f(t) = -1; \quad -\pi < t < 0$$

$$= 1; \quad 0 < t < \pi$$

————— ①

$$f(t) = f(t + 2\pi), \quad 2L = 2\pi$$

$$\therefore L = \pi$$

$$\therefore a_0 = \frac{1}{L} \int_{-\pi}^{\pi} f(t) dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) dt$$

$$= \frac{1}{\pi} \int_{-\pi}^0 f(t) dt + \frac{1}{\pi} \int_0^{\pi} f(t) dt$$

$$= \frac{1}{\pi} \int_{-\pi}^0 (-1) dt + \frac{1}{\pi} \int_0^{\pi} (1) dt \quad [\text{From (1)}]$$

$$= \frac{-1}{\pi} [t]_{-\pi}^0 + \frac{1}{\pi} [t]_0^{\pi}$$

$$= \frac{-1}{\pi} [0 - (-\pi)] + \frac{1}{\pi} [\pi - 0]$$

$$= -\frac{1}{\pi} [\pi] + \frac{1}{\pi} [\pi]$$

$$= -1 + 1$$

$$= 0$$

$$\therefore a_n = \frac{1}{L} \int_{-\pi}^{\pi} f(t) \cos \frac{n\pi t}{L} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos \frac{n\pi t}{\pi} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^0 f(t) \cos \frac{n\pi t}{\pi} dt + \frac{1}{\pi} \int_0^{\pi} f(t) \cos \frac{n\pi t}{\pi} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^0 (-1) \cos nt dt + \frac{1}{\pi} \int_0^{\pi} (1) \cos \frac{n\pi t}{\pi} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^0 (-1) \cos nt dt + \frac{1}{\pi} \int_0^{\pi} (1) \cos nt dt \quad [\text{From (1)}]$$

$$= -\frac{1}{\pi} \int_{-\pi}^0 \cos nt dt + \frac{1}{\pi} \int_0^{\pi} \cos nt dt$$

$$= -\frac{1}{\pi} \times \frac{1}{n} [\sin nt]_{-\pi}^0 + \frac{1}{\pi} \times \frac{1}{n} [\sin nt]_0^{\pi} \quad \left[\int \cos mx dx = \frac{1}{m} \sin mx \right]$$

$$= -\frac{1}{\pi n} [\sin 0 - \sin \frac{-n\pi}{\pi}] + \frac{1}{\pi n} [\sin \pi - 0]$$

$$= \frac{-1}{\pi n} [0 - \sin(-n\pi)] + \frac{1}{\pi n} [\sin n\pi - 0]$$

$$= -\frac{1}{\pi n} [0 + \sin(n\pi)] + \frac{1}{\pi n} [\sin n\pi - 0] \quad [\sin(-\theta) = -\sin\theta]$$

$$= -\frac{1}{\pi n} [0 + 0] + \frac{1}{\pi n} [0 - 0] \quad [\sin n\pi = 0 \text{ for } n = 1, 2, 3, \dots]$$

$$= 0$$

$$b_n = \frac{1}{L} \int_{-\pi}^{\pi} f(t) \sin \frac{n\pi t}{L} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin \frac{n\pi t}{\pi} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^0 f(t) \sin \frac{n\pi t}{\pi} dt + \frac{1}{\pi} \int_0^{\pi} f(t) \sin \frac{n\pi t}{\pi} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^0 -f(t) \sin nt dt + \frac{1}{\pi} \int_0^{\pi} f(t) \sin nt dt$$

$$= \frac{1}{\pi} \int_{-\pi}^0 (-1) \sin nt dt + \frac{1}{\pi} \int_0^{\pi} (1) \sin nt dt$$

$$= -\frac{1}{\pi} \int_{-\pi}^0 \sin nt dt + \frac{1}{\pi} \int_0^{\pi} \sin nt dt$$

$$= -\frac{1}{\pi} \times \frac{1}{n} [-\cos nt]_{-\pi}^0 + \frac{1}{\pi} \times \frac{1}{n} [-\cos nt]_0^{\pi}$$

$$= \frac{1}{\pi n} [\cos nt]_{-\pi}^0 + \frac{1}{\pi n} [-\cos nt]_0^{\pi}$$

$$= \frac{1}{\pi n} [\cos 0 - \cos(-n\pi)] - \frac{1}{\pi n} [\cos n\pi - \cos 0]$$

$$= \frac{1}{\pi n} [1 - \cos n\pi] - \frac{1}{\pi n} [\cos n\pi - 1]$$

$$= \frac{1}{\pi n} - \frac{1}{\pi n} \cos n\pi - \frac{1}{\pi n} \cos n\pi + \frac{1}{\pi n}$$

$$= \frac{2}{\pi n} - \frac{2}{\pi n} \cos n\pi$$

$$= \frac{2}{\pi n} (1 - \cos n\pi)$$

The Fourier series for the above function

$$\begin{aligned} f(t) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi t}{L} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi t}{L} \\ &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi t}{\pi} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi t}{\pi} \\ &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nt + \sum_{n=1}^{\infty} b_n \sin nt \\ &= 0 + \sum_{n=1}^{\infty} 0 \cdot \cos nt + \sum_{n=1}^{\infty} \frac{2}{n\pi} (1 - \cos n\pi) \sin nt \end{aligned}$$

$$\therefore \underbrace{f(t)}_{\text{Complex wave}} = \underbrace{0}_{\text{DC value}} + \underbrace{\sum_{n=1}^{\infty} \frac{2}{n\pi} (1 - \cos n\pi) \sin nt}_{\text{AC value.}}$$

(Ans)

(Ans)

Example - 24

$$y = f(t) = -t \quad ; \quad -\pi \leq t \leq 0$$

$$y = t \quad ; \quad 0 \leq t \leq \pi$$

$$f(t) = f(t + 2\pi) \quad , \quad T = 2\pi = 2\pi$$

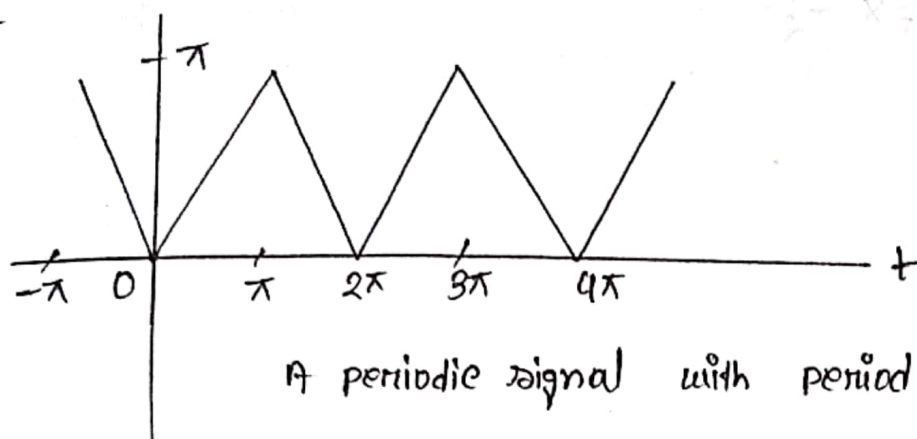
$$\therefore L = \pi$$

(a) sketch the function for 3 cycles.

(b) Find the fourier series for the function.

Solution:

(a)



A periodic signal with period $T = 2L = 2\pi$

(b) Given that,

$$y = -f(t) = -t; \quad -\pi \leq t \leq 0$$

$$y = t; \quad 0 \leq t \leq \pi \quad \text{--- ①}$$

$$f(t) = f(t + 2\pi)$$

$$\therefore T = 2L$$

$$= 2\pi$$

$$\therefore L = \pi.$$

$$\therefore a_0 = \frac{1}{L} \int_{-L}^L f(t) dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) dt$$

$$= \frac{1}{\pi} \int_{-\pi}^0 f(t) dt + \frac{1}{\pi} \int_0^{\pi} f(t) dt$$

$$= \frac{1}{\pi} \int_{-\pi}^0 (-t) dt + \frac{1}{\pi} \int_0^{\pi} (t) dt \quad [\text{From ①}]$$

$$= -\frac{1}{\pi} \left[\frac{t^2}{2} \right]_{-\pi}^0 + \frac{1}{\pi} \left[\frac{t^2}{2} \right]_0^{\pi} \quad \left[\int x^n dx = \frac{x^{n+1}}{n+1} + C \right]$$

$$= -\frac{1}{\pi} \left[0 - \frac{(-\pi)^2}{2} \right] + \frac{1}{\pi} \left[\frac{\pi^2}{2} - 0 \right]$$

$$\begin{aligned}
 &= -\frac{1}{\pi} \left[-\frac{\pi^2}{2} \right] + \frac{1}{\pi} \left[\frac{\pi^2}{2} \right] \\
 &= -\frac{1}{\pi} \times \left(-\frac{\pi^2}{2} \right) + \frac{1}{\pi} \times \frac{\pi^2}{2} \\
 &= \frac{\pi}{2} + \frac{\pi}{2} \\
 &= \frac{\pi + \pi}{2} \\
 &= \frac{2\pi}{2} \\
 &= \pi.
 \end{aligned}$$

$$\begin{aligned}
 \therefore a_n &= \frac{1}{L} \int_{-L}^L f(t) \cos \frac{n\pi t}{L} dt \\
 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos \frac{n\pi t}{\pi} dt \\
 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos(nt) dt \\
 &= \frac{1}{\pi} \int_{-\pi}^0 f(t) \cos(nt) dt + \frac{1}{\pi} \int_0^{\pi} f(t) \cos(nt) dt \\
 &= \frac{1}{\pi} \int_{-\pi}^0 (-t) \cos(nt) dt + \frac{1}{\pi} \int_0^{\pi} t \cos(nt) dt \quad [\text{From (1)}] \\
 &= -\frac{1}{\pi} \int_{-\pi}^0 t \cos(nt) dt + \frac{1}{\pi} \int_0^{\pi} t \cos(nt) dt \quad \text{--- (11)}
 \end{aligned}$$

Now, Let, $I = \int t \cos(nt) dt$

$$= t \int \cos(nt) dt - \int \left[\frac{d}{dt}(t) \int \cos(nt) dt \right] dt$$

$$\left[\because \int u v dx = u \int v dx - \int \left\{ \frac{d}{dx}(u) \int v dx \right\} dx \right]$$

$$= \frac{1}{n} \sin(nt) - \int \frac{1}{n} \sin(nt) dt \quad \left[\int \cos mx dx = \frac{1}{m} \sin mx \right]$$

$$= \frac{1}{n} \sin(nt) - \frac{1}{n} \int \sin(nt) dt$$

$$= \frac{1}{n} \sin(nt) - \frac{1}{n} \frac{-\cos(nt)}{n} \quad \left[\int \sin mx dx = -\frac{1}{m} \cos mx \right]$$

$$= \frac{1}{n} \sin(nt) + \frac{1}{n^2} \cos(nt) \quad \text{--- (iii)}$$

Putting the value of (iii) in equation (ii)

$$a_n = -\frac{1}{\pi} \int_{-\pi}^0 \frac{1}{n} \cos(nt) dt + \frac{1}{\pi} \int_0^{\pi} \frac{1}{n} \cos(nt) dt$$

$$= -\frac{1}{\pi} \left[\frac{1}{n} \sin(nt) + \frac{1}{n^2} \cos(nt) \right]_{-\pi}^0 + \frac{1}{\pi} \left[\frac{1}{n} \sin(nt) + \frac{1}{n^2} \cos(nt) \right]_0^{\pi}$$

$$= -\frac{1}{\pi} \left[\frac{0}{n} \sin(0) + \frac{1}{n^2} \cos(0) - \frac{(-\pi)}{n} \sin(-n\pi) - \frac{1}{n^2} \cos(-n\pi) \right]$$

$$+ \frac{1}{\pi} \left[\frac{\pi}{n} \sin(n\pi) + \frac{1}{n^2} \cos(n\pi) - \frac{0}{n} \sin(0) - \frac{1}{n^2} \cos(0) \right]$$

$$= -\frac{1}{\pi} \left[0 + \frac{1}{n^2} + \frac{\pi}{n} \sin(-n\pi) - \frac{1}{n^2} \cos(-n\pi) \right] + \frac{1}{\pi} \left[\frac{\pi}{n} \times 0 + \frac{1}{n^2} \cos(n\pi) - 0 - \frac{1}{n^2} \times 1 \right]$$

$$= -\frac{1}{\pi} \left[\frac{1}{n^2} - \frac{\pi}{n} \sin(n\pi) - \frac{1}{n^2} \cos(n\pi) \right] + \frac{1}{\pi} \left[\frac{1}{n^2} \cos(n\pi) - \frac{1}{n^2} \right]$$

$$\left[\sin(-\theta) = -\sin\theta; \cos(-\theta) = \cos\theta; \right. \\ \left. \sin n\pi = 0 \text{ for } n = 1, 2, 3, \dots \right]$$

$$= -\frac{1}{\pi} \left[\frac{1}{n^2} - 0 - \frac{1}{n^2} \cos(n\pi) \right] + \frac{1}{\pi} \left[\frac{1}{n^2} \cos(n\pi) - \frac{1}{n^2} \right]$$

$$= \frac{-1}{\pi n^2} + \frac{1}{\pi n^2} \cos(n\pi) + \frac{1}{\pi n^2} \cos(n\pi) - \frac{1}{\pi n^2}$$

$$= \frac{2}{\pi n^2} \cos(n\pi) - \frac{2}{\pi n^2}$$

$$= \frac{2}{\pi n^2} (\cos n\pi - 1).$$

$$\therefore a_n = \frac{2}{\pi n^2} (\cos n\pi - 1).$$

$$b_n = \frac{1}{L} \int_{-L}^L f(t) \sin \frac{n\pi t}{L} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin \frac{n\pi t}{\pi} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin(nt) dt$$

$$= \frac{1}{\pi} \int_{-\pi}^0 f(t) \sin(nt) dt + \frac{1}{\pi} \int_0^{\pi} f(t) \sin(nt) dt$$

$$= \frac{1}{\pi} \int_{-\pi}^0 f(-t) \sin(nt) dt + \frac{1}{\pi} \int_0^{\pi} t \sin(nt) dt$$

$$= -\frac{1}{\pi} \int_{-\pi}^0 t \sin(nt) dt + \frac{1}{\pi} \int_0^{\pi} t \sin(nt) dt \quad \text{--- (iv)}$$

Now,

$$\int t \sin(nt) dt$$

$$= t \int \sin(nt) dt - \int \left\{ \frac{d}{dt}(t) \int \sin(nt) dt \right\} dt$$

$$[\int u v dx = u \int v dx - \int \left\{ \frac{d}{dx}(u) \int v dx \right\} dx]$$

$$= t \left\{ \frac{-\cos(nt)}{n} \right\} - \int 1 \cdot \frac{-\cos(nt)}{n} dt$$

$$= -\frac{t}{n} \cos(nt) + \frac{1}{n} \int \cos(nt) dt$$

$$= -\frac{t}{n} \cos(nt) + \frac{1}{n} \frac{\sin(nt)}{n}$$

$$[\int \cos mx dx = \frac{1}{m} \sin mx]$$

$$= -\frac{t}{n} \cos(nt) + \frac{1}{n^2} \sin(nt) \quad \text{--- (v)}$$

Putting the value of (V) in equation (N)

$$\begin{aligned}
 b_n &= -\frac{1}{\pi} \int_{-\pi}^0 f(\sin nt) dt + \frac{1}{\pi} \int_0^{\pi} f(\sin nt) dt \\
 &= -\frac{1}{\pi} \left[-\frac{t}{n} \cos(nt) + \frac{1}{n^2} \sin(nt) \right]_{-\pi}^0 + \frac{1}{\pi} \left[-\frac{t}{n} \cos(nt) + \frac{1}{n^2} \sin(nt) \right]_0^{\pi} \\
 &= -\frac{1}{\pi} \left[\frac{-0}{n} \cos(0) + \frac{1}{n^2} \sin(0) - \frac{-(-\pi)}{n} \cos(-n\pi) - \frac{1}{n^2} \sin(-n\pi) \right] \\
 &\quad + \frac{1}{\pi} \left[-\frac{\pi}{n} \cos(n\pi) + \frac{1}{n^2} \sin(n\pi) - \frac{-0}{n} \cos(0) - \frac{1}{n^2} \sin(0) \right] \\
 &= -\frac{1}{\pi} \left[0+0 - \frac{\pi}{n} \cos(n\pi) + \frac{1}{n^2} \sin(n\pi) \right] + \frac{1}{\pi} \left[-\frac{\pi}{n} \cos(n\pi) + \frac{1}{n^2} \sin(n\pi) + 0-0 \right] \\
 &= -\frac{1}{\pi} \left[-\frac{\pi}{n} \cos(n\pi) + \frac{1}{n^2} \times 0 \right] + \frac{1}{\pi} \left[-\frac{\pi}{n} \cos(n\pi) + \frac{1}{n^2} \times 0 \right] \\
 &\quad \left[\sin(-\theta) = -\sin\theta; \cos(-\theta) = \cos\theta; \right. \\
 &\quad \left. \sin n\pi = 0 \text{ for } n = 1, 2, 3, \dots \right] \\
 &= \frac{\pi}{\pi n} \cos(n\pi) - \frac{\pi}{\pi n} \cos(n\pi) \\
 &= 0
 \end{aligned}$$

The fourier series for the above function is,

$$\begin{aligned}
 f(t) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi t}{L} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi t}{L} \\
 &= \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi n^2} (\cos n\pi - 1) + \sum_{n=1}^{\infty} 0 \cos \frac{n\pi t}{L} + 0 \\
 &= \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2 \pi} (\cos n\pi - 1) \cos \left(\frac{n\pi}{\pi} \right) + \\
 \therefore f(t) &= \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2 \pi} (\cos n\pi - 1) \cos nt
 \end{aligned}$$

complex
wave

DC
value

Ac value

(Ans)

Example - 25

$$y = f(t) = t ; -\pi \leq t \leq \pi$$

Here, $T = 2L = 2\pi$

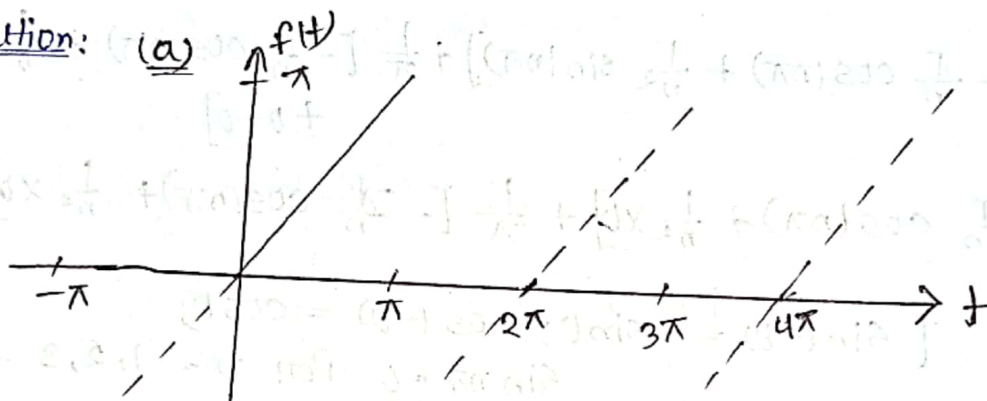
$$\therefore L = \pi$$

(a) Sketch the function for 3 cycles.

(b) Find the Fourier series for the function.

Solution:

(a)



A periodic signal with period $T = 2L = 2\pi$

$$\begin{aligned} \underline{(b)} \quad a_0 &= \frac{1}{L} \int_{-L}^L f(t) dt \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) dt \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} (t) dt \quad [\text{From (1)}] \\ &= \frac{1}{\pi} \left[\frac{t^2}{2} \right]_{-\pi}^{\pi} \\ &= \frac{1}{\pi} \left[\frac{\pi^2}{2} - \frac{(-\pi)^2}{2} \right] \\ &= \frac{1}{\pi} \left[\frac{\pi^2}{2} - \frac{\pi^2}{2} \right] \end{aligned}$$

$y = f(t) = t ; -\pi \leq t \leq \pi$
— (1)
 $L = \pi$

$$= 0$$

$$\begin{aligned} \therefore a_n &= \frac{1}{L} \int_{-L}^L f(t) \cos \frac{n\pi t}{L} dt \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos \frac{n\pi t}{\pi} dt \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos nt \, dt \quad [\text{From (1)}] \end{aligned}$$

$$\text{Now, } \int f(t) \cos nt \, dt$$

$$\begin{aligned} &= t \int \cos nt \, dt - \int \left\{ \frac{d}{dt}(t) \int \cos nt \, dt \right\} dt \\ &= \frac{t}{n} \times \sin nt - \int 1 \cdot \frac{\sin nt}{n} dt \quad \left[\because \int u v dx = u \int v dx - \int \left\{ \frac{d}{dx}(u) \int v dx \right\} dx \right] \\ &= \frac{t}{n} \sin nt - \frac{1}{n} \cdot \frac{1}{n} (-\cos nt) \\ &= \frac{t}{n} \sin nt + \frac{1}{n^2} \cos(nt) \quad \text{--- (11)} \end{aligned}$$

$$\begin{aligned} \therefore a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos nt \, dt \\ &= \frac{1}{\pi} \left[\frac{t}{n} \sin nt + \frac{1}{n^2} \cos nt \right]_{-\pi}^{\pi} \quad [\text{From (11)}] \\ &= \frac{1}{\pi} \left[\frac{\pi}{n} \sin n\pi + \frac{1}{n^2} \cos n\pi - \left\{ \frac{-\pi}{n} \sin(-n\pi) + \frac{1}{n^2} \cos(-n\pi) \right\} \right] \\ &= \frac{1}{\pi} \left[\frac{\pi}{n} \sin n\pi + \frac{1}{n^2} \cos n\pi + \frac{\pi}{n} \sin(-n\pi) - \frac{1}{n^2} \cos(-n\pi) \right] \\ &= \frac{1}{\pi} \left[\frac{\pi}{n} \sin n\pi + \frac{1}{n^2} \cos n\pi + \frac{\pi}{n} (-\sin n\pi) - \frac{1}{n^2} \cos(n\pi) \right] \\ &= \frac{1}{\pi} \left[\frac{\pi}{n} \sin n\pi + \frac{1}{n^2} \cos n\pi - \frac{\pi}{n} \sin n\pi - \frac{1}{n^2} \cos n\pi \right] \quad \left[\begin{array}{l} \sin(-\theta) = -\sin \theta; \\ \cos(-\theta) = \cos \theta \end{array} \right] \\ &= \frac{1}{\pi} \left[\frac{\pi}{n} \sin n\pi + \frac{1}{n^2} \cos n\pi - \frac{\pi}{n} \sin n\pi - \frac{1}{n^2} \cos n\pi \right] \end{aligned}$$

$$= \frac{1}{\pi} \times 0$$

$$= 0$$

$$\begin{aligned} \therefore b_n &= \frac{1}{L} \int_{-L}^L f(t) \sin \frac{n\pi t}{L} dt \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin \frac{n\pi t}{\pi} dt \quad [\text{From (1)}] \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin nt \, dt \end{aligned}$$

Now,

$$\begin{aligned} &\int f(t) \sin nt \, dt \\ &= t \int \sin nt \, dt - \int \left\{ \frac{d}{dt} (t) \int \sin nt \, dt \right\} dt \\ &\quad \left[\int u v dx = u \int v dx - \int \left\{ \frac{d}{dx} (u) \int v dx \right\} dx \right] \\ &= t \cdot \frac{1}{n} (-\cos nt) - \int 1 \cdot \frac{1}{n} (-\cos nt) dt \\ &= -\frac{t}{n} \cos nt + \frac{1}{n} \cos nt \, dt \\ &= -\frac{t}{n} \cos nt + \frac{1}{n^2} \sin nt \, dt \quad \text{--- (iii)} \\ &\quad \left[\int \cos mx = \frac{1}{m} \sin mx \right] \end{aligned}$$

$$\begin{aligned} \therefore b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin nt \, dt \\ &= \frac{1}{\pi} \left[-\frac{t}{n} \cos nt + \frac{1}{n^2} \sin nt \right]_{-\pi}^{\pi} \quad [\text{From (iii)}] \\ &= \frac{1}{\pi} \left[-\frac{\pi}{n} \cos n\pi + \frac{1}{n^2} \sin n\pi - \left\{ -\frac{(-\pi)}{n} \cos (-n\pi) + \frac{1}{n^2} \sin (-n\pi) \right\} \right] \end{aligned}$$

$$= \frac{1}{\pi} \left[-\frac{\pi}{n} \cos n\pi + \frac{1}{n^2} \sin n\pi - \left\{ \frac{\pi}{n} \cos(-n\pi) + \frac{1}{n^2} \sin(-n\pi) \right\} \right]$$

$$= \frac{1}{\pi} \left[-\frac{\pi}{n} \cos n\pi + \frac{1}{n^2} \sin n\pi - \frac{\pi}{n} \cos(-n\pi) - \frac{1}{n^2} (-) \sin(n\pi) \right]$$

$$[\sin(-\theta) = -\sin\theta; \cos(-\theta) = \cos\theta]$$

$$= \frac{1}{\pi} \left[-\frac{\pi}{n} \cos n\pi + \frac{1}{n^2} \sin n\pi - \frac{\pi}{n} \cos n\pi + \frac{1}{n^2} \sin n\pi \right]$$

$$= \frac{1}{\pi} \left[-\frac{\pi}{n} \cos n\pi - \frac{\pi}{n} \cos n\pi \right] \quad [\sin n\pi = 0]$$

$$\text{for } n = 1, 2, 3, \dots]$$

$$= -\frac{1}{\pi} \times \frac{2\pi}{n} \cos n\pi$$

$$= -\frac{2}{n} \cos n\pi$$

The Fourier series for the above function is,

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nt + \sum_{n=1}^{\infty} b_n \sin nt$$

$$= 0 + 0 + \sum_{n=1}^{\infty} b_n \sin nt$$

$$= \sum_{n=1}^{\infty} -\frac{2}{n} \cos n\pi \cdot \sin nt$$

$$\therefore f(t) = 0 + \left\{ -2 \sum_{n=1}^{\infty} \frac{1}{n} \cos n\pi \cdot \sin nt \right\}$$

Complex wave

DC Value

AC value

(Ans)

Example-31

If $y = f(t) = t^2$ over the interval $-\pi < t < \pi$ and has period 2π . Hence, $T = 2L = 2\pi \quad \therefore L = \pi$

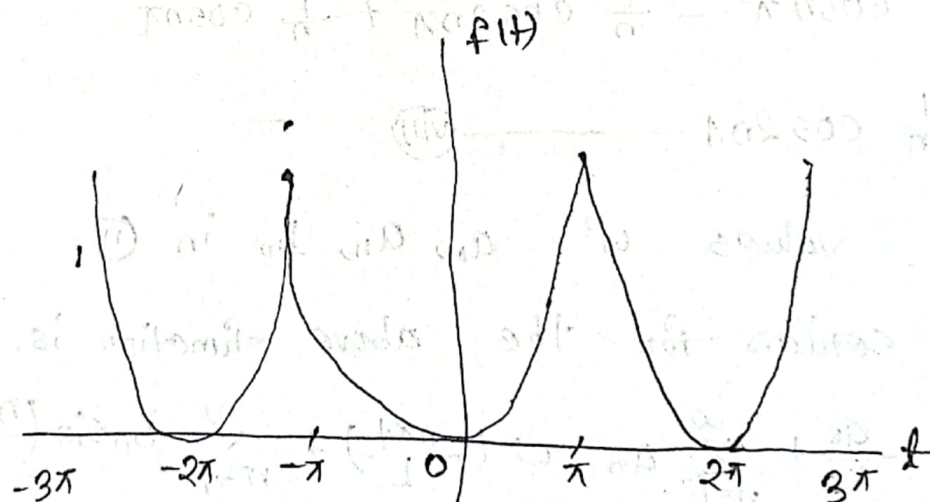
(a) sketch three cycles of $y = f(t)$ in the interval $-3\pi < t < 3\pi$

(b) Find the Fourier series for the function.

(c) Hence, Find the sum of the series $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$

Solution:

(a)



A periodic signal with period $T = 2L = 2\pi$

(b) $y = f(t) = t^2$, $-\pi < t < \pi$ — ①

$$T = 2L = 2\pi$$

$$\therefore L = \pi$$

$$a_0 = \frac{1}{L} \int_L^L f(t) dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} t^2 dt \quad [\text{From ①}]$$

$$= \frac{1}{\pi} \left[\frac{t^3}{3} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[\frac{\pi^3}{3} - \frac{(-\pi)^3}{3} \right]$$

$$= \frac{1}{\pi} \left[\frac{\pi^3}{3} + \frac{\pi^3}{3} \right]$$

$$= \frac{1}{\pi} \left[\frac{2\pi^3}{3} \right]$$

$$= \frac{2\pi^2}{3}$$

$$a_n = \frac{1}{L} \int_{-L}^L f(t) \cos \frac{n\pi t}{L} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos \frac{n\pi t}{\pi} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos nt dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} t^2 \cos nt dt \quad [\text{From ①}]$$

Now, $\int t^2 \cos nt \, dt$

$$= t^2 \int \cos nt \, dt - \int \left\{ \frac{d}{dt} (t^2) \int \cos nt \, dt \right\} dt$$

$$[\because \int u v dx = u \int v dx - \int \frac{d}{dx} (u) \int v dx dx]$$

$$= \frac{t^2}{n} \sin nt - \int 2t \cdot \frac{\sin nt}{n} dt$$

$$= \frac{t^2}{n} \sin nt - \frac{2}{n} \left[t \int \sin nt - \int \frac{d}{dt} (t) \int \sin nt dt \right] dt$$

$$= \frac{t^2}{n} \sin nt - \frac{2}{n} \left[t \cdot \frac{(-\cos nt)}{n} - \int \frac{(-\cos nt)}{n} dt \right]$$

$$= \frac{t^2}{n} \sin nt - \frac{2}{n} \left[\frac{-t \cos nt}{n} + \frac{\sin nt}{n^2} \right]$$

$$= \frac{t^2}{n} \sin nt + \frac{2t}{n^2} \cos nt - \frac{2}{n^3} \sin nt \quad \text{--- (11)}$$

$$\therefore a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} t^2 \cos nt \, dt$$

$$= \frac{1}{\pi} \left[\frac{t^2}{n} \sin nt + \frac{2t}{n^2} \cos nt - \frac{2}{n^3} \sin nt \right]_{-\pi}^{\pi} \quad [\text{From (11)}]$$

$$= \frac{1}{\pi} \left[\frac{\pi^2}{n} \sin n\pi + \frac{2\pi}{n^2} \cos n\pi - \frac{2}{n^3} \sin n\pi - \frac{(-\pi)^2}{n} \sin n(-\pi) - \frac{2(-\pi)}{n^2} \cos n(-\pi) + \frac{2}{n^3} \sin n(-\pi) \right]$$

$$= \frac{1}{\pi} \left[\frac{\pi^2}{n} \sin n\pi + \frac{2\pi}{n^2} \cos n\pi - \frac{2}{n^3} \sin n\pi + \frac{\pi^2}{n} \sin n\pi + \frac{2\pi}{n^2} \cos n\pi - \frac{2}{n^3} \sin n\pi \right]$$

$$[\sin(-\theta) = -\sin \theta; \cos(-\theta) = \cos \theta]$$

$$= \frac{1}{\pi} \left[0 + \frac{2\pi}{n^2} \cos n\pi - 0 + 0 + \frac{2\pi}{n^2} \cos n\pi - 0 \right] \quad [\sin n\pi = 0, \text{ For } n=1, 2, 3, \dots]$$

$$= \frac{1}{\pi} \left[\frac{2\pi}{n^2} \cos n\pi + \frac{2\pi}{n^2} \cos n\pi \right]$$

$$= \frac{1}{\pi} \left[\frac{4\pi}{n^2} \cos n\pi \right]$$

$$= \frac{4}{n^2} \cos n\pi$$

$$\therefore b_n = \frac{1}{L} \int_{-L}^L f(t) \sin \frac{n\pi t}{L} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \frac{\sin n\pi t}{\pi} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin nt dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} t^2 \sin nt dt \quad [\text{From (1)}]$$

Now, $\int t^2 \sin nt dt$

$$= t^2 \int \sin nt dt - \int \left\{ \frac{d}{dt} (t^2) \int \sin nt dt \right\} dt$$

$$[\because \int u v dx = u \int v dx - \int \left\{ \frac{d}{dx} (u) \int v dx \right\} dx]$$

$$= t^2 \int \sin nt dt - \int \left\{ 2t \cdot \frac{1}{n} (-\cos nt) \right\} dt$$

$$= t^2 \frac{1}{n} (-\cos nt) - \int \left\{ 2t \cdot \frac{1}{n} (-\cos nt) \right\} dt$$

$$= t^2 \frac{1}{n} (-\cos nt) + \frac{2}{n} \int t \cos nt dt$$

$$= t^2 \frac{1}{n} (-\cos nt) + \frac{2}{n} \left[t \int \cos nt dt - \int \left\{ \frac{d}{dt} (t) \int \cos nt dt \right\} dt \right]$$

$$= -t^2 \frac{1}{n} \cos nt + \frac{2}{n} \left[t \cdot \frac{1}{n} \sin nt - \int 1 \cdot \frac{1}{n} \sin nt dt \right]$$

$$\begin{aligned}
&= -t^2 \frac{1}{n} \cos nt + \frac{2}{n} \left[t \cdot \frac{1}{n} \sin nt - \frac{1}{n} \int \sin nt \, dt \right] \\
&= -t^2 \frac{1}{n} \cos nt + \frac{2}{n} \left[t \cdot \frac{1}{n} \sin nt - \frac{1}{n} \cdot \frac{1}{n} (-\cos nt) \right] \\
&= -t^2 \frac{1}{n} \cos nt + \frac{2}{n} \left[t \cdot \frac{1}{n} \sin nt + \frac{1}{n^2} \cos nt \right] \\
&= -t^2 \cdot \frac{1}{n} \cos nt + \frac{2t}{n^2} \sin nt + \frac{2}{n^3} \cos nt \quad \text{--- (11)}
\end{aligned}$$

$$\therefore b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} t^2 \sin nt \, dt$$

$$= \frac{1}{\pi} \left[-t^2 \frac{1}{n} \cos nt + \frac{2t}{n^2} \sin nt + \frac{2}{n^3} \cos nt \right]_{-\pi}^{\pi} \quad [\text{From (11)}]$$

$$\begin{aligned}
&= \frac{1}{\pi} \left[-\frac{\pi^2}{n} \cos n\pi + \frac{2\pi}{n^2} \sin n\pi + \frac{2}{n^3} \cos n\pi - \frac{-(-\pi)^2}{n} \cos n(-\pi) \right. \\
&\quad \left. - \frac{2(-\pi)}{n^2} \sin n(-\pi) - \frac{2}{n^3} \cos n(-\pi) \right]
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{\pi} \left[-\frac{\pi^2}{n} \cos n\pi + \frac{2\pi}{n^2} \sin n\pi + \frac{2}{n^3} \cos n\pi + \frac{\pi^2}{n} \cos n\pi \right. \\
&\quad \left. - \frac{2\pi}{n^2} \sin n\pi - \frac{2}{n^3} \cos n\pi \right]
\end{aligned}$$

$$[\sin(-\theta) = -\sin \theta; \cos(-\theta) = \cos \theta]$$

$$\begin{aligned}
&= \frac{1}{\pi} \left[-\frac{\pi^2}{n} \cos n\pi + 0 + \frac{2}{n^3} \cos n\pi + \frac{\pi^2}{n} \cos n\pi - 0 \right. \\
&\quad \left. - \frac{2}{n^3} \cos n\pi \right]
\end{aligned}$$

$$[\sin n\pi = 0; \text{ For } n = 1, 2, 3, \dots]$$

$$\begin{aligned}
&= \frac{1}{\pi} \left[-\frac{\pi^2}{n} \cos n\pi + \frac{2}{n^3} \cos n\pi + \frac{\pi^2}{n} \cos n\pi - \frac{2}{n^3} \cos n\pi \right] \\
&= \frac{1}{\pi} [0] = 0.
\end{aligned}$$

The Fourier series for the above function is,

$$\begin{aligned}
 \therefore f(t) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nt + \sum_{n=1}^{\infty} b_n \sin nt \\
 &= \frac{\frac{2\pi^2}{3}}{2} + \sum_{n=1}^{\infty} \frac{4}{n^2} \cos n\pi \cos nt + 0 \\
 &= \frac{\frac{2\pi^2}{3}}{2} + \sum_{n=1}^{\infty} \frac{4}{n^2} \cos n\pi \cos nt \\
 \therefore f(t) &= \frac{\frac{2\pi^2}{3}}{2} + \sum_{n=1}^{\infty} \frac{4}{n^2} \cos n\pi \cos nt \quad \text{--- (iv)}
 \end{aligned}$$

complex wave
DC Value
AC Value
(Ans)

(v)

$$f(t) = \frac{\frac{2\pi^2}{3}}{2} + \sum_{n=1}^{\infty} \frac{4}{n^2} \cos n\pi \cos nt$$

$$\Rightarrow f(t) = \frac{\pi^2}{3} + \left(\frac{4}{1^2} \cos \pi \cos t + \frac{4}{2^2} \cos 2\pi \cos 2t + \frac{4}{3^2} \cos 3\pi \cos 3t + \frac{4}{4^2} \cos 4\pi \cos 4t + \dots \right)$$

$$\Rightarrow f(t) = \frac{\pi^2}{3} + (-4 \cos t + \frac{4}{2^2} \cos 2t - \frac{4}{3^2} \cos 3t + \frac{4}{4^2} \cos 4t + \dots)$$

$$\Rightarrow f(t) = \frac{\pi^2}{3} - 4 \left(\cos t - \frac{1}{2^2} \cos 2t + \frac{1}{3^2} \cos 3t - \frac{1}{4^2} \cos 4t + \dots \right) \quad \text{--- (v)}$$

Given, $y = f(t) = t^2, \quad -\pi < t < \pi$

$$t^2 = \frac{\pi^2}{3} - 4 \left(\cos t - \frac{1}{2^2} \cos 2t + \frac{1}{3^2} \cos 3t - \frac{1}{4^2} \cos 4t + \dots \right) \quad \text{--- (vi)}$$

Put $t=0$ in (i)

$$0^2 = \frac{\pi^2}{3} - 4 \left(\cos 0 - \frac{1}{2^2} \cos 2 \cdot 0 + \frac{1}{3^2} \cos 3 \cdot 0 - \frac{1}{4^2} \cos 4 \cdot 0 + \dots \right)$$

$$\Rightarrow 0^2 = \frac{\pi^2}{3} - 4 \left(1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots \right)$$

$$\Rightarrow -\frac{\pi^2}{3} = -4 \left(1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots \right)$$

$$\Rightarrow \frac{\pi^2}{3} = 4 \left(1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots \right)$$

$$\Rightarrow \frac{\pi^2}{12} = \left(1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots \right)$$

$$\therefore 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}$$

(Ans)

Problem-19 (Derive complex form of Fourier series)

we have,

The trigonometric form of Fourier series,

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\omega t) + \sum_{n=1}^{\infty} b_n \sin(n\omega t) \quad \text{--- (1)}$$

$$\text{we have} = 1 + \frac{x^1}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \quad \text{--- (2)}$$

Putting $x = ix$

$$e^{ix} = 1 + \frac{ix}{1!} + \frac{(ix)^2}{2!} + \frac{(ix)^3}{3!} + \dots$$

$$\text{on, } e^{ix} = 1 + \frac{ix}{1!} - \frac{x^2}{2!} - \frac{ix^3}{3!} + \frac{x^4}{4!} + \frac{ix^5}{5!} - \frac{x^6}{6!} + \dots$$

$$\text{on, } e^{ix} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots + \left(\frac{ix}{1!} - \frac{ix^3}{3!} + \frac{ix^5}{5!} - \frac{ix^7}{7!} + \dots \right)$$

$$\text{on, } e^{ix} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots + i \left(\frac{x}{1!} - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \right)$$

$$\text{on, } e^{ix} = \cos x + i \sin x \quad \text{--- (3)}$$

Putting $x = -ix$

$$e^{-ix} = 1 + \frac{-ix}{1!} + \frac{(-ix)^2}{2!} + \frac{(-ix)^3}{3!} + \frac{(-ix)^4}{4!} + \dots$$

$$\text{on, } e^{-ix} = 1 - \frac{ix}{1!} - \frac{x^2}{2!} + \frac{ix^3}{3!} + \frac{x^4}{4!} - \frac{ix^5}{5!} - \frac{x^6}{6!} + \frac{ix^7}{7!} + \dots$$

$$\text{on, } e^{-ix} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots + (-i) \left(\frac{x}{1!} - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \right)$$

$$\text{on, } e^{-ix} = \cos x - i \sin x \quad \text{--- (4)}$$

Putting (3) and (4)

$$e^{ix} + e^{-ix} = 2 \cos x$$

$$\text{on, } \cos x = \frac{1}{2} (e^{ix} + e^{-ix}) \quad \text{--- (5)}$$

Subtracting (4) from (5)

$$e^{ix} - e^{-ix} = 2i \sin x$$

$$\text{or, } \sin x = \frac{1}{2i} (e^{ix} - e^{-ix}) \quad \text{--- (6)}$$

From (5),

$$\cos x = \frac{1}{2} (e^{ix} + e^{-ix})$$

$$\text{Hence, } \cos(n\omega t) = \frac{1}{2} (e^{in\omega t} + e^{-in\omega t}) \quad \text{--- (7)}$$

From (6)

$$\sin x = \frac{1}{2i} (e^{ix} - e^{-ix})$$

$$\text{Hence, } \sin(n\omega t) = \frac{1}{2i} (e^{in\omega t} - e^{-in\omega t}) \quad \text{--- (8)}$$

Putting The values of (7) and (8) in equation (1)

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\omega t) + \sum_{n=1}^{\infty} b_n \sin(n\omega t)$$

$$= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \left\{ \frac{1}{2} (e^{in\omega t} + e^{-in\omega t}) \right\} + \sum_{n=1}^{\infty} b_n \left\{ \frac{1}{2i} (e^{in\omega t} - e^{-in\omega t}) \right\}$$

$$= \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \left\{ \frac{1}{2} (e^{in\omega t} + e^{-in\omega t}) \right\} + b_n \left\{ \frac{1}{2i} (e^{in\omega t} - e^{-in\omega t}) \right\}]$$

$$= \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[\frac{1}{2} a_n e^{in\omega t} + \frac{1}{2i} b_n e^{in\omega t} + \frac{1}{2} a_n e^{-in\omega t} + \frac{1}{2i} (-b_n e^{-in\omega t}) \right]$$

$$= \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[\frac{1}{2} a_n e^{in\omega t} + \frac{(-1)(-1)}{2i} b_n e^{in\omega t} + \frac{1}{2} a_n e^{-in\omega t} + \frac{(-1)(-1)}{2i} (-b_n e^{-in\omega t}) \right]$$

$$= \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[\frac{1}{2} (a_n e^{in\omega t}) + \frac{(-1)i}{2} (b_n e^{in\omega t}) + \frac{1}{2} (a_n e^{-in\omega t}) + \frac{(-1)i}{2} (-b_n e^{-in\omega t}) \right]$$

$$= \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[\left\{ \frac{1}{2} (a_n e^{in\omega t}) - \frac{i}{2} (b_n e^{in\omega t}) \right\} + \left\{ \frac{1}{2} (a_n e^{-in\omega t}) + \frac{i}{2} (b_n e^{-in\omega t}) \right\} \right]$$

$$= \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[\frac{1}{2} (a_n - ib_n) e^{in\omega t} + \frac{1}{2} (a_n + ib_n) e^{-in\omega t} \right]$$

$$\text{Let, } c_0 = \frac{a_0}{2} \quad \text{--- (9)}$$

$$c_n = \frac{1}{2} (a_n - ib_n) \quad \text{--- (10)}$$

$$c_{-n}^* = \frac{1}{2} (a_n + ib_n) \quad \text{--- (11)}$$

Then,

$$f(t) = c_0 + \sum_{n=1}^{\infty} [c_n e^{in\omega t} + c_{-n}^* e^{-in\omega t}]$$

$$= c_0 + \sum_{n=1}^{\infty} [c_n e^{in\omega t} + c_{-n} e^{-in\omega t}]$$

$$= c_0 + \sum_{n=1}^{\infty} c_n e^{in\omega t} + \sum_{n=1}^{\infty} c_{-n} e^{-in\omega t}$$

$$= c_0 + \sum_{n=1}^{\infty} c_n e^{in\omega t} + \sum_{n=-1}^{-\infty} c_n e^{in\omega t}$$

$$= c_0 + \sum_{n=-\infty}^{\infty} c_n e^{in\omega t} \quad \text{--- (12)}$$

[when, $n=0$, $c_0=1$]

Hence, complex form of Fourier series:

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{in\omega t}$$

$$\text{where, } c_n = \frac{1}{2} (a_n - ib_n)$$

Convolution sum: The following steps are:

1. Folding
2. Shifting
3. Multiplication
4. Summation

1st time: 1. Folding 2. Multiplication 3. Summation

2nd time: 1. Shifting 2. Multiplication 3. Summation

Example: 102

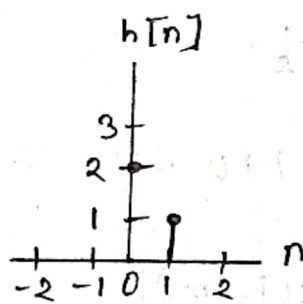
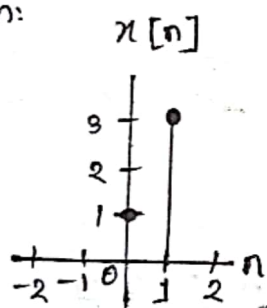
Evaluate the convolution sums of $y[n] = x[n] * h[n]$

where, $x[n] = 1, n=0$
 $= 3, n=1$

and $h[n] = 2, n=0$
 $= 1, n=1$

n represents the time index.

Solution:



Matlab

$x = [1 \ 3]$

$h = [2 \ 1]$

$y = \text{conv}(x, h)$

$y = [2 \ 7 \ 3]$

1st time:

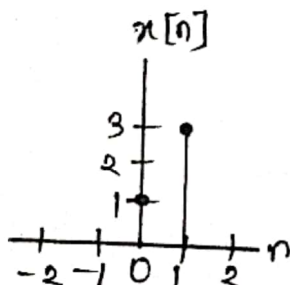
(i) Folding:

$$\begin{aligned} x[n] &= x[-n] \\ x[0] &= x[0] \\ x[1] &= x[-1] \end{aligned}$$

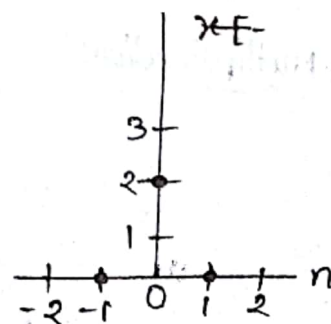
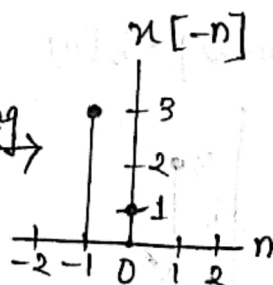
(ii)

Multiplication:

$$x[-n] * h[n]$$



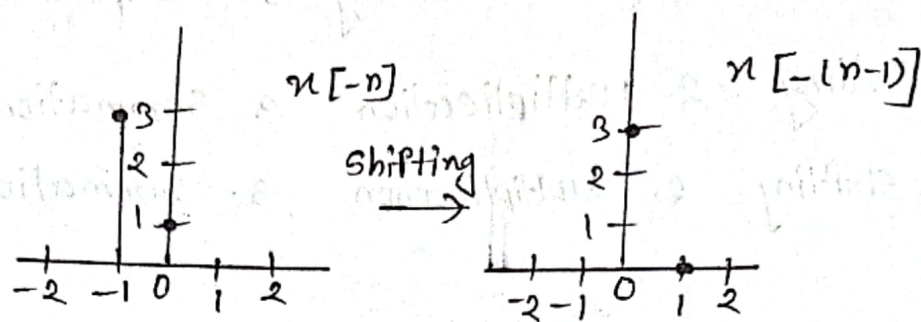
Folding $\rightarrow$



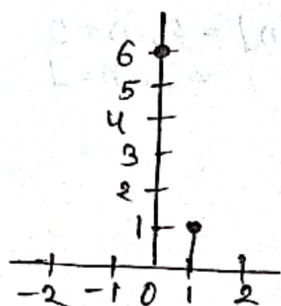
(iii)

$$\begin{aligned} \text{Summation: } y[0] &= 0 + 2 + 0 \\ &= 2 \end{aligned}$$

2nd time: (i) Shifting:

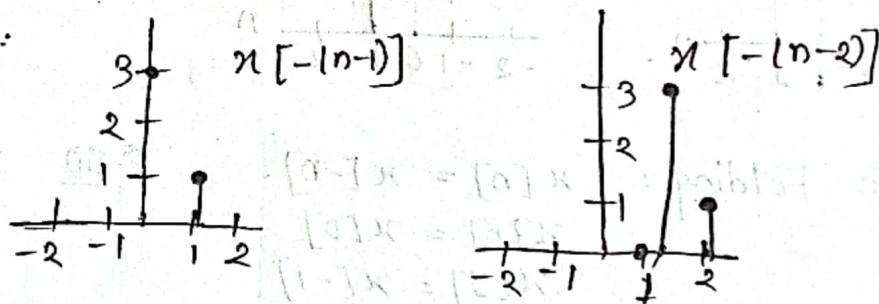


(ii) Multiplication: $x[-(n-1)] * h[n]$

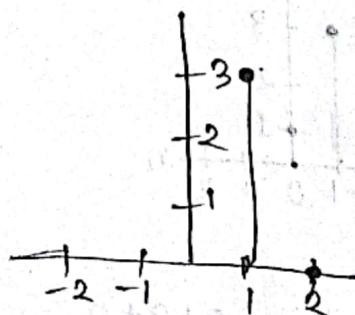


(iii) Summation: $y[1] = 0 + 6 + 1 + 0 = 7$

3rd time: (i) Shifting:

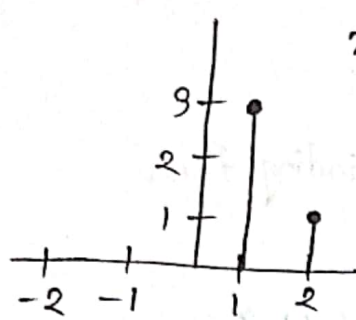


(ii) Multiplication: $x[-(n-2)] * h[n]$

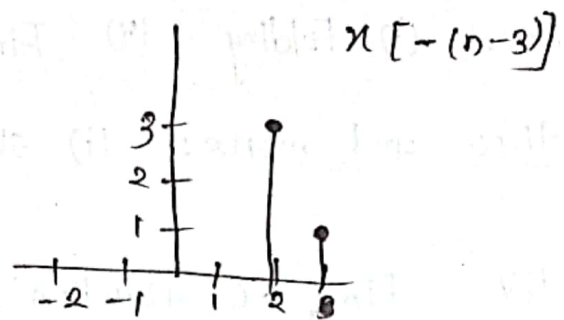


(iii) Summation: $y[2] = 0 + 0 + 3 + 0 = 3$

4th time: (i) Shifting:

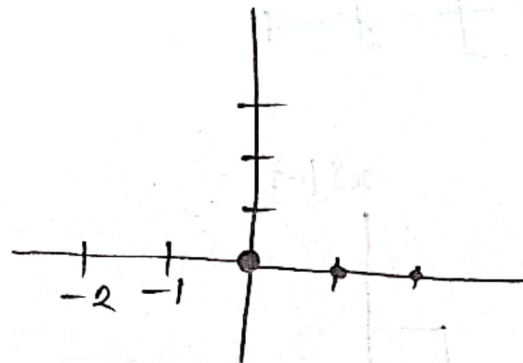


Shifting $\rightarrow$



(ii) Multiplication:

$$x[-(n-3)] * h[n]$$



(iii) Summation:

$$y[3] = 0 + 0 + 0 + 0 = 0$$

Finally we get,

$$y[0] = 2$$

$$y[1] = 7$$

$$y[2] = 3$$

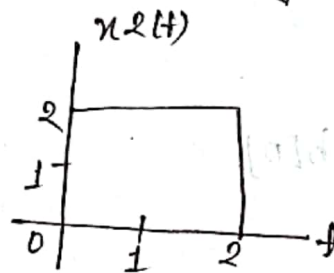
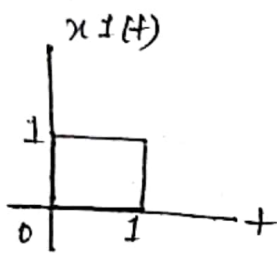
$\therefore$ Convolution sum: $y[n] = [2 \ 7 \ 3]$.

Convolution integral: (i) Folding (ii) shifting (iii) Finding Area

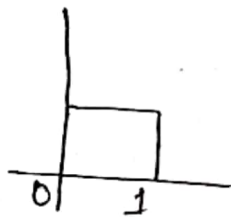
1st time: (i) Folding (ii) Finding Area

2nd time and more: (i) shifting (ii) Finding Area

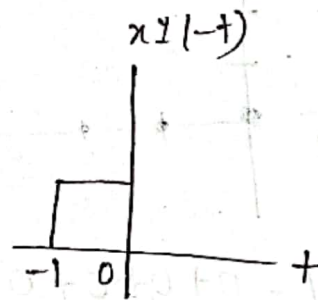
Example - 107 Find convolution integral of $x_1(t) * x_2(t)$



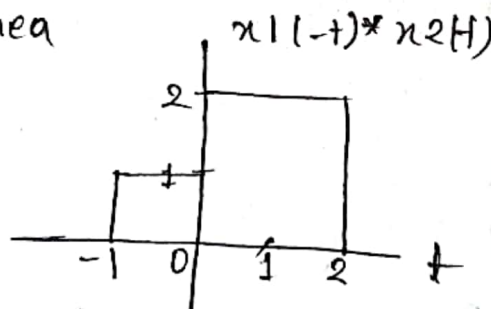
1st time: (i) Folding



Folding
→



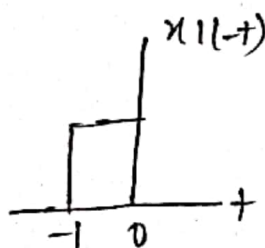
(ii) Finding Area



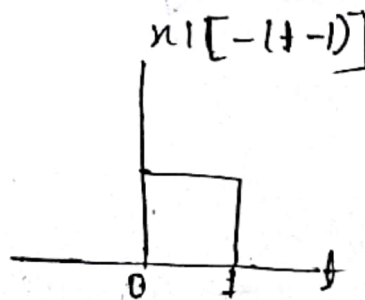
No overlapped area; $x[0] = 0$

2nd time:

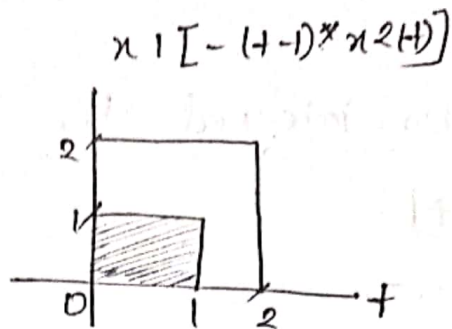
(i) shifting



shifting
→

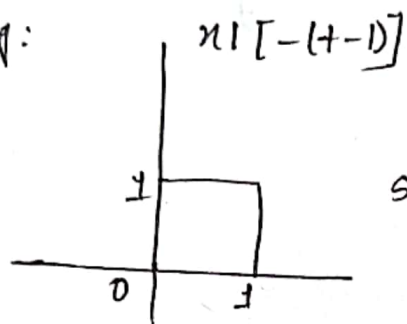


(ii) Finding Area

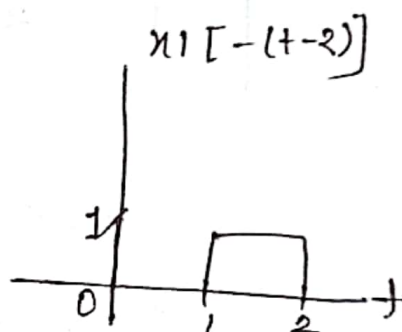


Area: $x[1] = 1 \times 1 = 1$

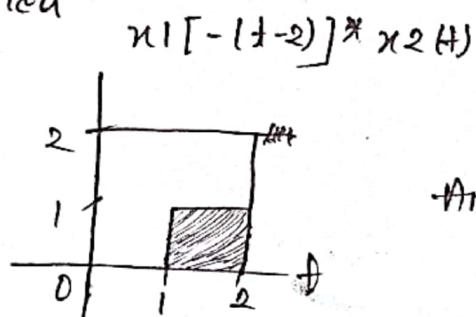
(iii) Shifting:



shifting

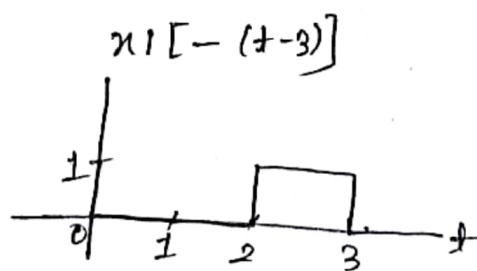


(iv) Finding Area

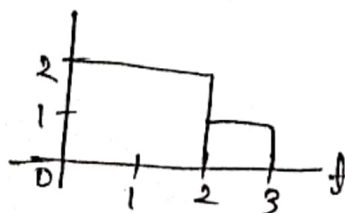


Area: $x[2] = 1 \times 1 = 1$

v Shifting:

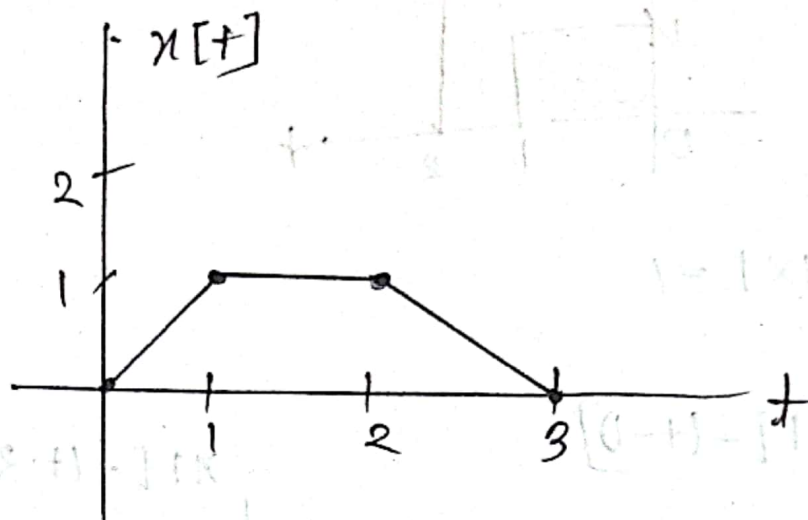


(vi) Finding Area: $x_1[-(t-3)] * x_2(t)$



No overlapped: $x[3] = 0$

hence the convolution integral is,



(Ans)

Example-58

Find Laplace transform of $f(t) = e^{at}$

Solution: Given that,

$$f(t) = e^{at}$$

we know that,

$$L\{f(t)\} = \int_0^{\infty} f(t) e^{-st} dt$$

$$= \int_0^{\infty} e^{at} e^{-st} dt$$

$$= \int_0^{\infty} e^{at-st} dt$$

$$= \int_0^{\infty} e^{-st+at} dt$$

$$= \int_0^{\infty} e^{-(s-a)t} dt$$

$$= \int_0^{\infty} e^{-(s-a)t} dt$$

$$= \left[\frac{e^{-(s-a)t}}{-(s-a)} \right]_0^{\infty}$$

$$= \frac{1}{-(s-a)} [e^{-\infty} - e^{-0}]$$

$$= \frac{-1}{(s-a)} \left[\frac{1}{e^{\infty}} - \frac{1}{e^0} \right]$$

$$= -\frac{1}{(s-a)} \left[\frac{1}{\infty} - \frac{1}{1} \right]$$

$$= -\frac{1}{(s-a)} [0 - 1]$$

$$= \frac{1}{s-a}$$

(Ans)

$$\left[\int e^{-mx} dx = \frac{e^{-mx}}{-m} \right]$$

$$[e^{\infty} = \infty]$$

$$\left[\frac{1}{\infty} = 0 \right]$$

Example: 60

Find Laplace Transform of $f(t) = \sin at$.

Solution: Given that,

$$f(t) = \sin at$$

We know that,

$$L\{f(t)\} = \int_0^{\infty} f(t) \cdot e^{-st} dt$$

$$\text{Hence,} \quad = \int_0^{\infty} (\sin at) \cdot e^{-st} dt \quad \text{--- (1)}$$

Again, we know that,

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$\therefore e^{iat} = \cos at + i\sin at$$

Hence, $\cos at$ is the real (R) part of e^{iat} and

$\sin at$ is the imaginary (I) part of e^{iat}

That is $\cos at = R(e^{iat})$ and $\sin at = I(e^{iat})$

From (1) we have,

$$L(\sin at) = \int_0^{\infty} \sin at \cdot e^{-st} dt$$

$$= \int_0^{\infty} I e^{iat} \cdot e^{-st} dt$$

$$= I \int_0^{\infty} e^{(ia-s)t} dt$$

$$= I \left\{ \left[\frac{e^{-(s-ia)t}}{-(s-ia)} \right]_0^{\infty} \right\}$$

$$= I \left\{ \frac{e^{-(s-ia)\infty}}{-(s-ia)} - \frac{e^{-(s-ia) \cdot 0}}{-(s-ia)} \right\}$$

$$= \mathcal{I} \left\{ 0 - \frac{e^{-0}}{-(s-ia)} \right\}$$

$$= \mathcal{I} \left[\frac{1}{s-ia} \right]$$

$$= \mathcal{I} \left\{ \frac{s+ia}{(s-ia)(s+ia)} \right\}$$

$$= \mathcal{I} \left\{ \frac{s+ia}{s^2 - i^2 a^2} \right\}$$

$$= \mathcal{I} \left\{ \frac{s+ia}{s^2 + a^2} \right\} [i^2 = -1]$$

$$= \mathcal{I} \left\{ \frac{s}{s^2 + a^2} + i \frac{a}{s^2 + a^2} \right\} \text{--- (ii)}$$

Taking that imaginary part from (ii)

$$L(f(t)) = L(\sin at)$$

$$= \frac{a}{s^2 + a^2} \text{ (Ans)}$$

Example: 61 Find Laplace Transform of $f(t) = \cos at$

Solution: Given that, $f(t) = \cos at$

We know that,

$$\mathcal{L}\{f(t)\} = \int_0^\infty f(t) e^{-st} dt \text{--- (i)}$$

$$= \int_0^\infty \cos at \cdot e^{-st} dt \text{--- (ii)}$$

Again, we know that,

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$\therefore e^{iat} = \cos at + i \sin at$$

Here, $\cos at$ is the real part of e^{iat} and $\sin at$ is the imaginary part of e^{iat} .

That is, $\cos at = R(e^{iat})$ and $\sin at = I(e^{iat})$

[$R(e^{iat})$ means real part of e^{iat} and $I(e^{iat})$ means imaginary part of e^{iat}]

are

From (ii) we have,

$$L(\cos at) = \int_0^{\infty} \cos at \cdot e^{-st} dt$$

$$= \int_0^{\infty} R(e^{iat}) \cdot e^{-st} dt$$

$$= R \int_0^{\infty} e^{iat} \cdot e^{-st} dt$$

$$= R \int_0^{\infty} e^{(ia-s)t} dt$$

$$= R \int_0^{\infty} e^{-(s-ia)t} dt$$

$$= R \left\{ \left[\frac{e^{-(s-ia)t}}{-(s-ia)} \right]_0^{\infty} \right\}$$

$$= R \left\{ \frac{e^{-(s-ia)\infty}}{-(s-ia)} - \frac{e^{-(s-ia) \cdot 0}}{-(s-ia)} \right\}$$

$$= R \left\{ 0 - \frac{e^{-0}}{-(s-ia)} \right\}$$

$$= R \left[\frac{1}{s-ia} \right]$$

$$[\because e^{-\infty} = \frac{1}{e^{\infty}} = \frac{1}{\infty} = 0]$$

$$[e^{-0} = \frac{1}{e^0} = \frac{1}{1} = 1]$$

$$= R \left\{ \frac{stia}{(stia)(s-ia)} \right\}$$

$$= R \left\{ \frac{stia}{s^2 - i^2 a^2} \right\}$$

$$= R \left\{ \frac{stia}{s^2 + a^2} \right\} [i^2 = -1]$$

$$\therefore R \left\{ \frac{s}{s^2+a^2} + i \frac{a}{s^2+a^2} \right\} \text{ --- (iv)}$$

Taking that real part from (iv)

$$L\{f(t)\} = L(\cos at) = \frac{s}{s^2+a^2}$$

(Ans)

□ As for example,

$$L(\sin at) = \frac{a}{s^2+a^2}$$

$$\therefore L(\sin 5t) = \frac{5}{s^2+5^2}$$

$$L(\cos at) = \frac{s}{s^2+a^2}$$

$$\therefore L(\sin 3t) = \frac{3}{s^2+3^2}$$

The 1st shift theorem

$$\mathcal{L}\{f(t)\} = f(s)$$

The 1st shift theorem states that,

$$\text{If } \mathcal{L}\{f(t)\} = f(s) \quad \text{--- (I)}$$

$$\text{Then } \mathcal{L}\{e^{-at}f(t)\} = f(s+a) \quad \text{--- (II)}$$

Example-64

$$\text{Find } \mathcal{L}\{e^{-at}\} = ?$$

OR, state First shift theorem using the theorem
find the values of $\mathcal{L}\{e^{-at}\} = ?$

Solution, The first shift theorem state that,

$$\text{if } \mathcal{L}\{f(t)\} = f(s) \text{ then } \mathcal{L}\{e^{-at}f(t)\} = f(s+a) \quad \text{--- (I)}$$

According to the question,

$$\text{Let, } f(t) = t$$

$$\text{Now, } \mathcal{L}\{f(t)\} = \mathcal{L}\{t\} = \frac{1}{s^2} \quad [\text{Hence, } f(s) = \frac{1}{s^2}]$$

$$\text{If } f(s) = \frac{1}{s^2}$$

$$\therefore f(s+a) = \frac{1}{(s+a)^2}$$

According to equation (I)

$$\mathcal{L}\{e^{-at}t\} = f(s+a) = \frac{1}{(s+a)^2} \quad \text{(Ans)}$$

Example: 68

Find $L \{ e^{-3t} * e^{5t} \} = ?$

Here, $f(t) = e^{5t}$

The 1st shift theorem states that,

$$\text{If } L \{ f(t) \} = f(s) \quad \text{--- (I)}$$

$$\text{Then } L \{ e^{-3t} f(t) \} = f(s+3) \quad \text{--- (II)}$$

According to equation (I)

$$L \{ f(t) \} = L \{ e^{5t} \} = \frac{1}{s-5} \quad \left[f(s) = \frac{1}{s-5} \right]$$

$$\text{If } f(s) = \frac{1}{s-5}$$

$$\therefore f(s+3) = \frac{1}{s+3-5}$$

According to equation no (II) we can write,

$$L \{ e^{-3t} * e^{5t} \} = f(s+3)$$

$$= \frac{1}{s+3-5}$$

$$= \frac{1}{(s-2)}$$

(Ans)

Example - 72

Find $L\{t \cdot e^{4t}\} = ?$

Solution: Here, $f(t) = e^{4t}$

This theorem states that,

$$\text{If } L\{f(t)\} = f(s) \text{ ——— (i)}$$

$$\text{Then } L\{t \cdot f(t)\} = -\frac{d}{ds} \{f(s)\} \text{ ——— (ii)}$$

According to equation no (i)

$$L\{f(t)\} = L\{e^{4t}\} = \frac{1}{s-4} \quad [\text{Here, } f(s) = \frac{1}{s-4}]$$

Hence, According to equation no (ii) we can write

$$L\{t \cdot f(t)\} = -\frac{d}{ds} \{f(s)\}$$

$$L\{t \cdot e^{4t}\} = -\frac{d}{ds} \{f(s)\}$$

$$[f(t) = e^{4t}]$$

$$L\{t \cdot e^{4t}\} = -\frac{d}{ds} \left\{ \frac{1}{s-4} \right\}$$

$$[f(s) = \frac{1}{s-4}]$$

$$= -\frac{d}{ds} (s-4)^{-1}$$

$$= -(-1)(s-4)^{-1-1} \frac{d}{ds} (s-4)$$

$$= (s-4)^{-2} (1-0)$$

$$= (s-4)^{-2}$$

$$= \frac{1}{(s-4)^2}$$

$$\therefore L\{t \cdot e^{4t}\} = \frac{1}{(s-4)^2}$$

(Ans)

Division Theorem

Division theorem states that,

$$\mathcal{L}\{f(t)\} = f(s)$$

$$\text{Then } \mathcal{L}\left[\frac{1}{t} f(t)\right] = \int_s^\infty f(s) ds$$

Example-75 Find laplace transform of $\left\{\frac{\sin 2t}{t}\right\}$

Solution: Here, $f(t) = \sin 2t$

Division theorem states that,

$$\mathcal{L}\{f(t)\} = f(s) \quad \text{--- ①}$$

$$\mathcal{L}\left\{\frac{1}{t} f(t)\right\} = \int_s^\infty f(s) ds \quad \text{--- ②}$$

According to ①,

$$\mathcal{L}\{f(t)\} = \mathcal{L}(\sin 2t)$$

$$[f(s) = \frac{2}{s^2 + 4}]$$

According to division theorem,

$$f(s) = \frac{2}{s^2 + 4}$$

$$\Rightarrow \mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty f(s) ds$$

$$\begin{aligned} \Rightarrow \mathcal{L}\left\{\frac{\sin 2t}{t}\right\} &= \int_s^\infty \frac{2}{s^2 + 4} ds \\ &= 2 \int_s^\infty \frac{ds}{s^2 + 4} \end{aligned}$$

$$= 2 \cdot \frac{1}{2} [\tan^{-1} \frac{s}{2}]$$

$$= \tan^{-1} \infty - \tan^{-1} \frac{s}{2}$$

$$= \pi/2 - \tan^{-1} \frac{s}{2}$$

$$\therefore \int \frac{\sin 2t}{2} = \pi/2 - \tan^{-1} \frac{s}{2}.$$

(Ans)

Example: 101 Find inverse Laplace transform of $\frac{s+4}{s(s-1)(s-2)}$

Solution: Let,

$$\frac{s+4}{s(s-1)(s-2)} = f(s)$$

$$\frac{s+4}{s(s-1)(s-2)} \equiv \frac{A}{s} + \frac{B}{(s-1)} + \frac{C}{(s-2)} \quad \text{--- (i)}$$

$$\text{OR, } s+4 = A(s-1)(s-2) + Bs(s-2) + Cs(s-1) \quad \text{--- (ii)}$$

Now, $s=0$ in equation (ii)

$$0+4 = A(0-1)(0-2) + B \cdot 0(0-2) + C \cdot 0(0-1)$$

$$\Rightarrow 4 = 2A$$

$$\Rightarrow A = 2$$

$\therefore$ put $s-1=0$; i.e. $s=1$ in equation (ii)

$$1+4 = A(2-1)(1-2) + B \cdot 1(1-2) + C \cdot 1(1-1)$$

$$\Rightarrow 5 = 0 - B + 0$$

$$B = -5$$

Put, $s-2=0$; i.e. $s=2$ in equation (1)

$$\Rightarrow 2+4 = A(2-1)(2-2) + B \times 2(2-2) + C \times 2(2-1)$$

$$\Rightarrow 6 = 0 + 0 + 2C$$

$$\therefore C = 3$$

Putting the value of A, B, C in equation (1)
we get,

$$\frac{s+4}{s(s-1)(s-2)} = \frac{2}{s} + \frac{-5}{s-1} + \frac{3}{s-2}$$

$$\text{on, } L^{-1} \left\{ \frac{s+4}{s(s-1)(s-2)} \right\} = L^{-1} \left(\frac{2}{s} \right) + L^{-1} \left(\frac{-5}{s-1} \right) + L^{-1} \left(\frac{3}{s-2} \right)$$

$$\text{on, } L^{-1} \left\{ \frac{s+4}{s(s-1)(s-2)} \right\} = 2L^{-1} \left(\frac{1}{s} \right) - 5L^{-1} \left(\frac{1}{s-1} \right) + 3L^{-1} \left(\frac{1}{s-2} \right) \quad \text{--- (11)}$$

we know that,

$$L(f(t)) = L(1) = \frac{1}{s}$$

$$\Rightarrow 1 = L^{-1} \left(\frac{1}{s} \right)$$

$$\therefore L^{-1} \left(\frac{1}{s} \right) = 1$$

Again,

$$\therefore L(f(t)) = L(e^{at}) = \frac{1}{s-a}$$

$$\Rightarrow e^{at} = L^{-1} \left(\frac{1}{s-a} \right)$$

$$\Rightarrow e^t = L^{-1} \left(\frac{1}{s-1} \right)$$

$$\therefore L^{-1} \left(\frac{1}{s-1} \right) = e^t.$$

$$\therefore L(f(t)) = L(e^{at}) = \frac{1}{s-a}$$

$$\Rightarrow e^{at} = L^{-1} \frac{1}{s-a}$$

$$\Rightarrow e^{2t} = L^{-1} \left(\frac{1}{s-2} \right)$$

$$\therefore L^{-1} \left(\frac{1}{s-2} \right) = e^{2t}$$

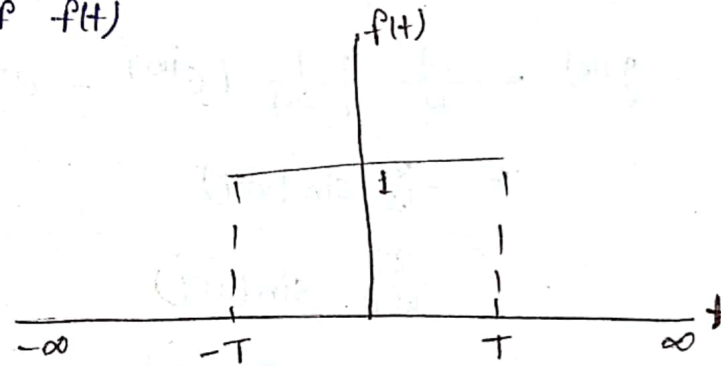
Putting the values in (ii) we get,

$$\begin{aligned} L^{-1} \left(\frac{s+4}{s(s-1)(s-2)} \right) &= 2L^{-1} \left(\frac{1}{s} \right) - 5L^{-1} \left(\frac{1}{s-1} \right) + 3L^{-1} \left(\frac{1}{s-2} \right) \\ &= 2 \cdot 1 - 5e^t + 3e^{2t} \\ &= 2 - 5e^t + 3e^{2t} \end{aligned}$$

(Ans)

39. Consider the rectangular pulse figure 84 described as,
 $f(t) = 1 ; -T \leq t \leq T$
 $= 0 ; |t| > T$

Find the Fourier transform of $f(t)$



Solution:

we have,

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} [g(\omega)] e^{i\omega t} d\omega$$

where, $g(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$

$$= \int_{-\infty}^{-T} f(t) e^{-i\omega t} dt + \int_{-T}^T f(t) e^{-i\omega t} dt + \int_T^{\infty} f(t) e^{-i\omega t} dt$$

$$= \int_{-\infty}^{-T} 0 \cdot e^{-i\omega t} dt + \int_{-T}^T 1 \cdot e^{-i\omega t} dt + \int_T^{\infty} 0 \cdot e^{-i\omega t} dt$$

$$= \int_{-T}^T 1 \cdot e^{-i\omega t} dt$$

$$= \left[\frac{e^{-i\omega t}}{-i\omega} \right]_{-T}^T$$

$$\left[\int e^{-mx} dx = \frac{e^{-mx}}{-m} \right]$$

$$= \frac{-1}{i\omega} (e^{-i\omega T} - e^{i\omega T})$$

$$= \frac{1}{i\omega} (e^{i\omega T} - e^{-i\omega T})$$

$$g(\omega) = \frac{2 \cdot 1}{\omega} \left\{ \frac{1}{2i} (e^{i\omega T} - e^{-i\omega T}) \right\}$$

$$= \frac{2}{\omega} \sin(\omega T) \quad \text{--- (i)}$$

$$\left[\because \sin x = \frac{1}{2i} (e^{ix} - e^{-ix}) \right] \text{ [pdf-4, pg-84, equation (viii)]}$$

$$\left[\text{Since we have, } \sin x = \frac{1}{2i} (e^{ix} - e^{-ix}) \right]$$

$$g(\omega) = \frac{2 \cdot 1}{\omega} \left\{ \frac{1}{2i} (e^{i\omega T} - e^{-i\omega T}) \right\}$$

$$= \frac{2}{\omega} \sin(\omega T)$$

$$= \frac{2T}{\omega T} \sin(\omega T)$$

$$g(\omega) = 2T \frac{\sin(\omega T)}{\omega T}$$

$$\text{Thus, we write for all } \omega, g(\omega) = 2T \frac{\sin(\omega T)}{\omega T} \quad \text{--- (1)}$$

Both (i) and (ii) are correct.

For $\omega = 0$, the integral simplifies to $2T$. It is straight-forward to show using L'Hopital's rule that, From (i)

$$\lim_{\omega \rightarrow 0} \frac{2}{\omega} \sin(\omega T)$$

$$= \lim_{\omega \rightarrow 0} \frac{2 \sin(\omega T)}{\omega} \quad \left[\frac{0}{0}, \text{ in determinate form} \right]$$

$$= \lim_{\omega \rightarrow 0} \frac{2 \cos(\omega T) \cdot T}{1} \quad \left[\text{differentiate numerator and denominator with respect to } \omega \right]$$

$$= \frac{2 \cos 0 \cdot T}{1}$$

$$= 2 \cdot 1 \cdot T = 2T$$

Again, from (ii),

$$g(\omega) = 2T \frac{\sin(\omega T)}{\omega T}$$

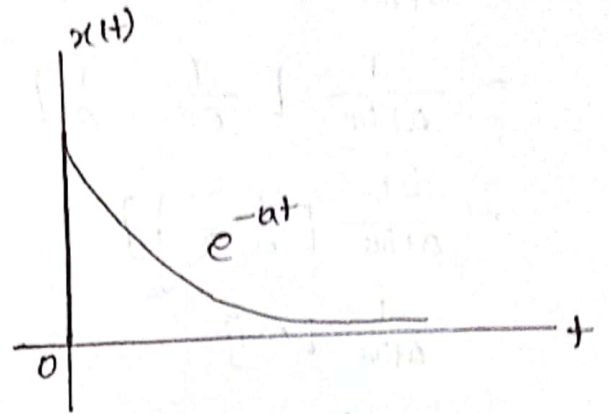
$$\lim_{\omega \rightarrow 0} g(\omega) = \lim_{\omega \rightarrow 0} 2T \frac{\sin(\omega T)}{\omega T}$$

$$\lim_{\omega \rightarrow 0} g(\omega) = 2T \frac{\sin(\omega T)}{\omega T}$$

$$\lim_{\omega \rightarrow 0} g(\omega) = 2T \cdot 1 \quad \left[\because \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1 \right]$$

Example: 40. Find the Fourier transform of $f(t) = e^{-at} u(t)$

Solution: The Fourier transform does not converge for $a \leq 0$, since $f(t)$ is not absolutely integrable as shown by:



(i) For $a > 0$, we have,

$$g(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

$$= \int_{-\infty}^{\infty} e^{-at} u(t) e^{-i\omega t} dt \quad [f(t) = e^{-at} u(t)]$$

$$= \int_{-\infty}^0 e^{-at} u(t) e^{-i\omega t} dt + \int_0^{\infty} e^{-at} u(t) e^{-i\omega t} dt$$

$$= \int_{-\infty}^0 e^{-at} \cdot 0 \cdot e^{-i\omega t} dt + \int_0^{\infty} e^{-at} \cdot 1 \cdot e^{-i\omega t} dt$$

[By definition from unit step function]

$$= 0 + \int_0^{\infty} e^{-at} \cdot 1 \cdot e^{-i\omega t} dt$$

$$= \int_0^{\infty} e^{-at} e^{-i\omega t} dt$$

$$= \int_0^{\infty} e^{-at} e^{-i\omega t} dt$$

$$= \int_0^{\infty} e^{-(a+i\omega)t} dt \quad \text{--- ①}$$

$$= \frac{-1}{a+i\omega} \left[e^{-(a+i\omega)t} \right]_0^{\infty}$$

$$= \frac{-1}{a+i\omega} \left[e^{-(a+i\omega)\infty} - e^{-(a+i\omega)0} \right]$$

$$= \frac{-1}{a+i\omega} [e^{-\infty} - e^{-0}]$$

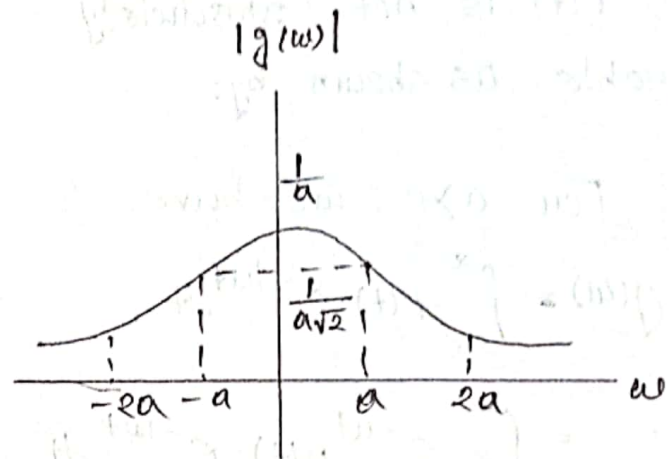
$$= \frac{-1}{a+i\omega} \left[\frac{1}{e^{\infty}} - \frac{1}{e^0} \right] [e^{\infty} - \infty]$$

$$= \frac{-1}{a+i\omega} \left[\frac{1}{\infty} - \frac{1}{1} \right]$$

$$= \frac{-1}{a+i\omega} [0-1]$$

$$= \frac{1}{a+i\omega}$$

$$\therefore |g(\omega)| = \frac{1}{\sqrt{a^2 + \omega^2}}$$



(ii) for $a < 0$

$$\therefore g(\omega) = \int_0^{\infty} e^{-(a+i\omega)t} dt$$

$$= \int_0^{\infty} e^{-at - i\omega t} dt$$

$$= \int_0^{\infty} e^{at - i\omega t} dt$$

$$= \int_0^{\infty} e^{(a-i\omega)t} dt$$

$$= \frac{1}{a-i\omega} [e^{(a-i\omega)t}]_0^{\infty}$$

$$= \frac{1}{a-i\omega} [e^{(a-i\omega)\infty} - e^{(a-i\omega)0}]_0^{\infty}$$

$$= \frac{1}{a-i\omega} [e^{\infty} - e^0]$$

$$= \frac{1}{a-i\omega} [\infty - 1]$$

$$= \infty$$

$f(t)$ on Magnitude spectrum

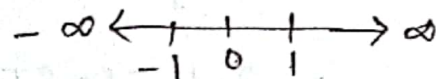
(Ans)

Example - 41 Find Fourier transform of

$$f(t) = 1 \quad ; \quad 0 \leq t < 1$$

$$= -1 \quad ; \quad -1 \leq t < 0$$

$$= 0 \quad ; \quad |t| > 1 \quad / \text{ OR } t > 1$$



Solution: We know that,

$$g(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

$$\therefore g(\omega) = \int_{-\infty}^{-1} f(t) e^{-i\omega t} dt + \int_{-1}^0 f(t) \cdot e^{-i\omega t} dt + \int_0^1 f(t) e^{-i\omega t} dt + \int_1^{\infty} f(t) e^{-i\omega t} dt$$

$$= \int_{-\infty}^{-1} 0 \cdot e^{-i\omega t} dt + \int_{-1}^0 (-1) e^{-i\omega t} dt + \int_0^1 1 e^{-i\omega t} dt + \int_1^{\infty} 0 \cdot e^{-i\omega t} dt$$

$$= - \int_{-1}^0 e^{-i\omega t} dt + \int_0^1 e^{-i\omega t} dt$$

$$= - \left[\frac{e^{-i\omega t}}{-i\omega} \right]_{-1}^0 + \left[\frac{e^{-i\omega t}}{-i\omega} \right]_0^1 \quad \left[\because \int e^{-mx} dx = \frac{e^{-mx}}{-m} \right]$$

$$= - \left[\frac{e^{-i\omega \cdot 0}}{-i\omega} - \frac{e^{-i\omega(-1)}}{-i\omega} \right] + \left[\frac{e^{-i\omega \cdot 1}}{-i\omega} - \frac{e^{-i\omega \cdot 0}}{-i\omega} \right]$$

$$= - \left[\frac{e^{-0}}{-i\omega} - \frac{e^{i\omega}}{-i\omega} \right] + \left[\frac{e^{-i\omega}}{-i\omega} - \frac{e^{-0}}{-i\omega} \right]$$

$$= - \left[\frac{1}{-i\omega} - \frac{e^{i\omega}}{-i\omega} \right] + \left[\frac{e^{-i\omega}}{-i\omega} - \frac{1}{-i\omega} \right]$$

$$= - \left[\frac{1}{-i\omega} - \frac{e^{i\omega}}{-i\omega} \right] + \left[\frac{e^{-i\omega}}{-i\omega} - \frac{1}{-i\omega} \right] \quad [e^0 = 1]$$

$$= - \frac{1}{-i\omega} + \frac{e^{i\omega}}{-i\omega} + \frac{e^{-i\omega}}{-i\omega} - \frac{1}{-i\omega}$$

$$= \left[\frac{1}{i\omega} - \frac{e^{i\omega}}{i\omega} \right]$$

$$= \frac{1}{i\omega} + \frac{1}{i\omega} - \frac{e^{i\omega}}{i\omega} - \frac{e^{-i\omega}}{i\omega}$$

$$= \frac{2}{i\omega} - \frac{1}{i\omega} (e^{i\omega} + e^{-i\omega})$$

$$= \frac{2}{i\omega} - \frac{1}{i\omega} 2 \cos \omega$$

$$= \frac{2}{i\omega} (1 - \cos \omega)$$

$$\left[\begin{aligned} \frac{e^{i\theta} + e^{-i\theta}}{2} &= \cos \theta \\ \therefore e^{i\theta} + e^{-i\theta} &= 2 \cos \theta \end{aligned} \right]$$

(Ans)

Example: 43 Find Fourier transform for the given functions.

$$f(t) = e^{-2t} ; t \geq 0$$

$$f(t) = 0 ; t < 0$$

Solution: Given that,

$$f(t) = e^{-2t} ; t \geq 0$$

$$f(t) = 0 ; t < 0$$

We know that, $g(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$

$$= \int_{-\infty}^0 f(t) e^{-i\omega t} dt + \int_0^{\infty} f(t) e^{-i\omega t} dt$$

$$= \int_{-\infty}^0 0 \cdot e^{-i\omega t} dt + \int_0^{\infty} e^{-2t} e^{-i\omega t} dt$$

$$= \int_0^{\infty} e^{-2t} e^{-i\omega t} dt$$

$$= \int_0^{\infty} e^{-2t - i\omega t} dt$$

$$= \int_0^{\infty} e^{-(2+i\omega)t} dt$$

$$= \left[\frac{e^{-(2+i\omega)t}}{-(2+i\omega)} \right]_0^{\infty}$$

$$= \frac{-1}{2+i\omega} [e^{-\infty} - e^{-0}]$$

$$= \frac{-1}{2+i\omega} \left[\frac{1}{e^{\infty}} - \frac{1}{e^0} \right]$$

$$= \frac{-1}{2+i\omega} \left[\frac{1}{\infty} - \frac{1}{1} \right]$$

$$= \frac{-1}{2+i\omega} [0 - 1] = \frac{1}{2+i\omega}$$

$$\therefore |g(\omega)| = \frac{1}{\sqrt{2^2 + \omega^2}}$$

$$[i^2 = -1]$$

$$= \frac{1}{\sqrt{4 + \omega^2}}$$

(Ans)

Example: 89

$$x(t) = -u(t+3) + 2u(t+1) - 2u(t-1) + u(t-3)$$

Solution: Given that,

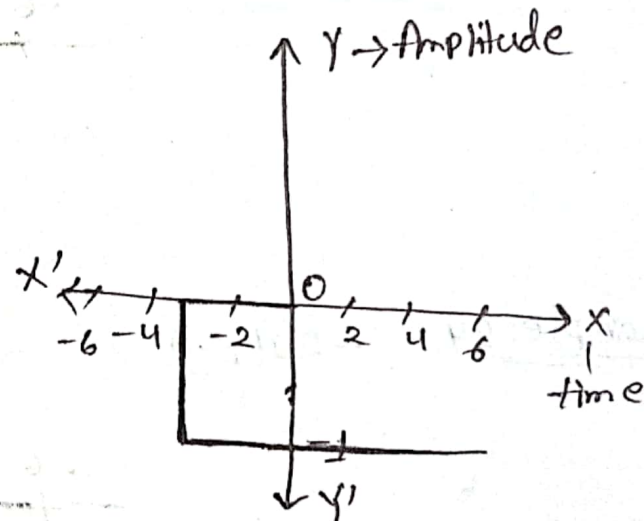
$$x(t) = -u(t+3) + 2u(t+1) - 2u(t-1) + u(t-3)$$

Now, for

$-u(t+3)$ we have,

$$-u(t+3) = \begin{cases} 1, & t+3 \geq 0 \\ 0, & t+3 < 0 \end{cases}$$

$$= \begin{cases} 1, & t \geq -3 \\ 0, & t < -3 \end{cases}$$

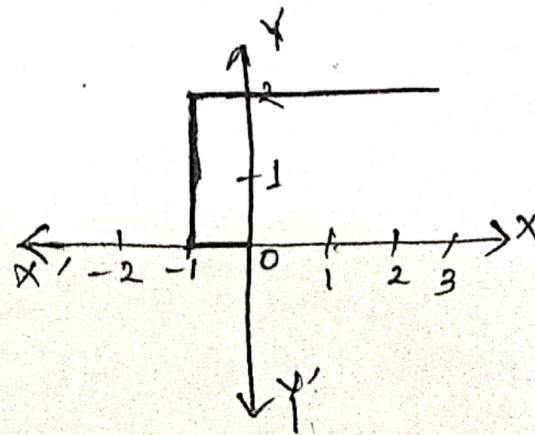


Again, for,

$2u(t+1)$ we have,

$$2u(t+1) = \begin{cases} 2, & t+1 \geq 0 \\ 0, & t+1 < 0 \end{cases}$$

$$= \begin{cases} 2, & t \geq -1 \\ 0, & t < -1 \end{cases}$$

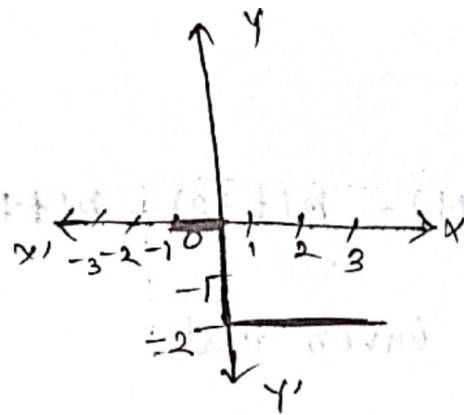


Again for,

$-2U(t-1)$ we have,

$$-2U(t-1) = \begin{cases} -2, & t-1 \geq 0 \\ 0, & t-1 < 0 \end{cases}$$

$$= \begin{cases} -2, & t \geq 1 \\ 0, & t < 1 \end{cases}$$

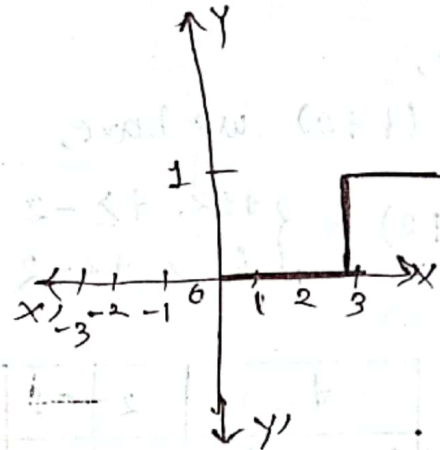


Again for,

$U(t-3)$ we have,

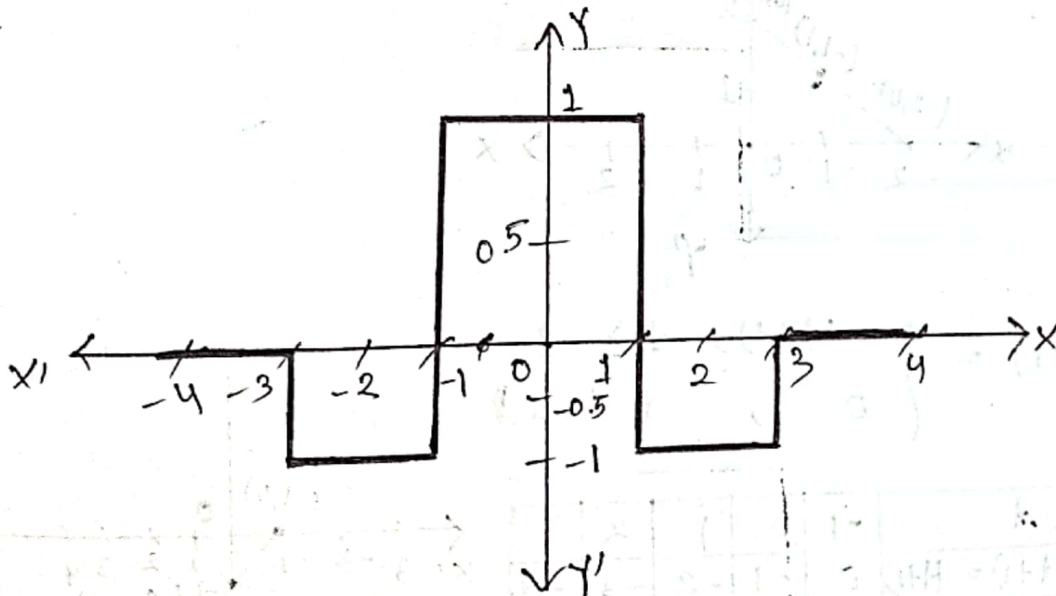
$$U(t-3) = \begin{cases} 1, & t-3 \geq 0 \\ 0, & t-3 < 0 \end{cases}$$

$$= \begin{cases} 1, & t \geq 3 \\ 0, & t < 3 \end{cases}$$



Finally for,

$$x(t) = -U(t+3) + 2U(t+1) - 2U(t-1) + U(t-3)$$



Example: 91 Find $\mathcal{L}[u(t-a)] = \frac{e^{-as}}{s}$

Solution: we have,

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} f(t) e^{-st} dt$$

Here, $f(t) = u(t-a)$

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} f(t) e^{-st} dt$$

$$\therefore \mathcal{L}\{u(t-a)\} = \int_0^{\infty} u(t-a) e^{-st} dt$$

$$= \int_0^a 0 \cdot e^{-st} dt + \int_a^{\infty} 1 \cdot e^{-st} dt$$

$$= \int_0^a 0$$

$$= 0 + \int_a^{\infty} 1 \cdot e^{-st} dt$$

$$= \int_a^{\infty} e^{-st} dt$$

$$= \left[\frac{e^{-st}}{-s} \right]_a^{\infty}$$

$$= -\frac{1}{s} [e^{-s \times \infty} - e^{-s \times a}]$$

$$= -\frac{1}{s} [e^{-\infty} - e^{-as}]$$

$$= -\frac{1}{s} \left[\frac{1}{e^{\infty}} - e^{-as} \right]$$

$$= -\frac{1}{s} \left[\frac{1}{\infty} - e^{-as} \right]$$

$$= -\frac{1}{s} [0 - e^{-as}]$$

$$= +\frac{1}{s} [e^{-as}]$$

$$= \frac{e^{-as}}{s}$$

$$\therefore L\{v(t-a)\} = \frac{e^{-as}}{s} \quad [\text{Proved}]$$

$$\therefore L\{v(t-2)\} = \frac{e^{-2s}}{s}$$

Example-92

Express the following function in terms of unit step functions and find its laplace transform:

$$f(t) = \begin{cases} 8; & t < 2 \\ 6; & t > 2 \end{cases}$$

Solution: Given that,

$$f(t) = \begin{cases} 8; & t < 2 \\ 6; & t > 2 \end{cases}$$

$$\text{we have, } L\{v(t-a)\} = \frac{e^{-as}}{s}$$

$$\therefore L\{v(t-2)\} = \frac{e^{-2s}}{s} \quad \text{--- (1)}$$

$$\text{Now, } f(t) = \begin{cases} 8; & t < 2 \\ 6; & t > 2 \end{cases}$$

$$= \begin{cases} 8+0; & t < 2 \\ 8-2; & t > 2 \end{cases}$$

$$= 8 + \begin{cases} 0; & t < 2 \\ -2; & t > 2 \end{cases}$$

$$= 8 + (-2) \begin{cases} 0; & t < 2 \\ 1; & t > 2 \end{cases}$$

$$= 8 + (-2) \begin{cases} 1; & t > 2 \\ 0; & t < 2 \end{cases}$$

$$\therefore f(t) = 8 + (t-2) u(t-2) \\ = 8 - 2u(t-2)$$

$$\therefore \mathcal{L}\{f(t)\} = \mathcal{L}\{8 - 2u(t-2)\} \\ = \mathcal{L}\{8\} - 2\mathcal{L}\{u(t-2)\} \\ = 8\mathcal{L}\{1\} - 2\mathcal{L}\{u(t-2)\} \\ = 8 \times \frac{1}{s} - 2 \times \frac{e^{-2s}}{s} \quad \left[\mathcal{L}\{1\} = \frac{1}{s}; \right. \\ \left. \mathcal{L}\{u(t-2)\} = \frac{e^{-2s}}{s} \right] \\ \text{(Ans)}$$

Example-90

$$x(t) = h(t+2) - h(t+1) - h(t-1) + h(t-2)$$

Solution: Given that,

$$x(t) = h(t+2) - h(t+1) - h(t-1) + h(t-2)$$

Now, for,

$h(t+2)$ we have,

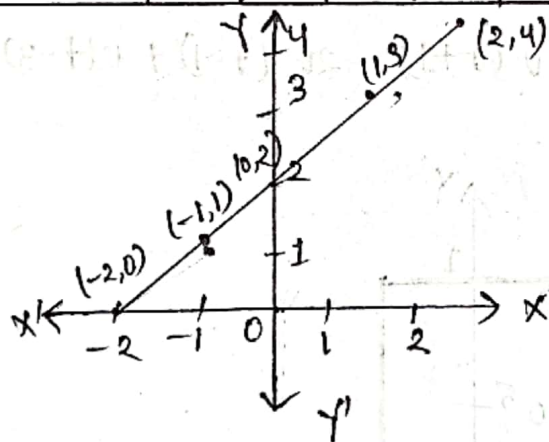
$$h(t+2) = \begin{cases} t+2, & t \geq -2 \\ 0, & t < -2 \end{cases}$$

Hence,

$$t+2 \geq 0$$

$$\therefore t \geq -2$$

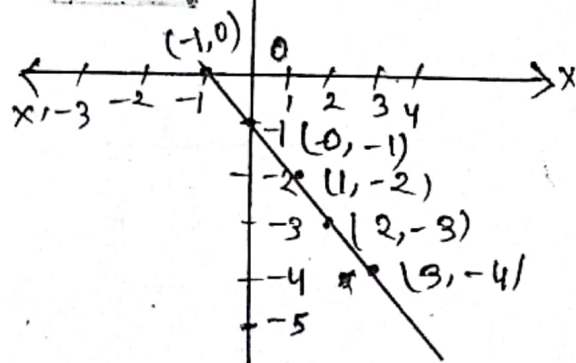
t	-2	-1	0	1	2
$h(t+2) = t+2$	0	1	2	3	4



Again,

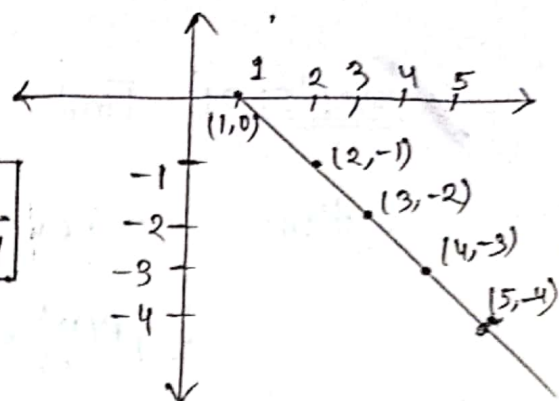
$$-h(t+1) = \begin{cases} -(t+1), & t \geq -1 \\ 0, & t < -1 \end{cases}$$

t	-1	0	1	2	3
$-h(t+1) = -(t+1)$	0	-1	-2	-3	-4



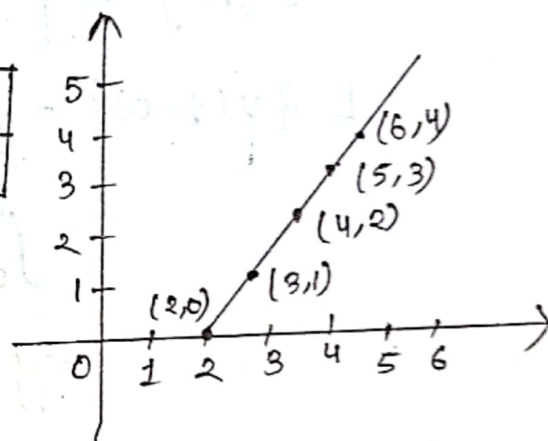
$$\therefore -\pi(t-1) = \begin{cases} -(t-1) & ; t \geq 1 \\ 0 & ; t < 1 \end{cases}$$

t	1	2	3	4	5
$-\pi(t-1)$	0	-1	-2	-3	-4

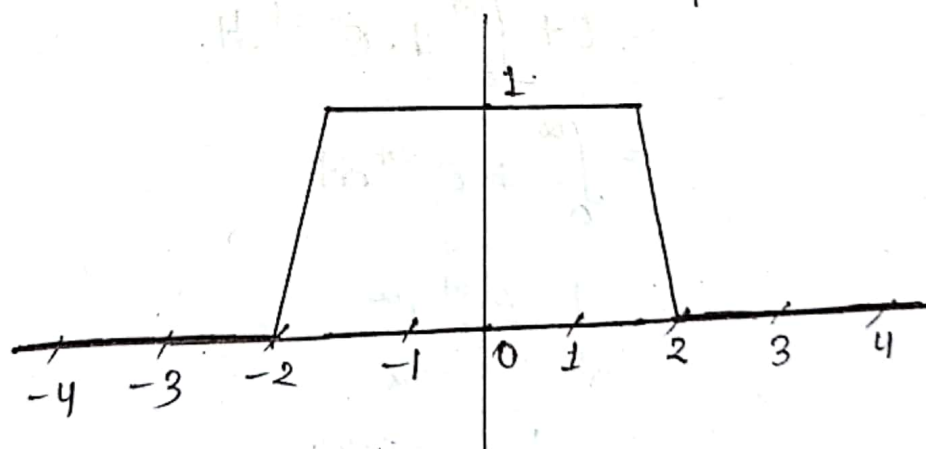


$$\therefore \pi(t-2) = \begin{cases} t-2 & ; t \geq 2 \\ 0 & ; t < 2 \end{cases}$$

t	2	3	4	5	6
$\pi(t-2) = t-2$	0	1	2	3	4



Final for,



103Impulse differential equation using Laplace transform:103

Solve the following initial value problem (IVP) by Laplace Transform $Y'' + Y = t$, $Y(0) = 1$, $Y'(0) = -2$.

Solution:

$$\text{Let, } Y = f(t)$$

$$Y' = f'(t)$$

$$Y'' = f''(t)$$

$$\text{Given that, } Y'' + Y = t$$

$$\text{That is, } \frac{d^2 Y}{dt^2} + Y = t$$

Taking the Laplace transform of both sides of the differential equation and using the given conditions,

we have,

$$Y'' + Y = t$$

$$L\{Y''\} + L\{Y\} = L\{t\}$$

we have,

$$L\{f''(t)\} = s^2 L\{f(t)\} - sf(0) - f'(0)$$

$$L\{Y''\} + L\{Y\} = L\{t\}$$

$$\Rightarrow s^2 L\{Y\} - sf(0) - f'(0) + L\{Y\} = L\{t\}$$

$$\Rightarrow s^2 Y - sf(0) - f'(0) + Y = L\{t\} \quad \left[\begin{array}{l} \text{Let, } L\{Y\} = Y \\ \text{Let, } L\{t\} = \frac{1}{s^2} \end{array} \right]$$

$$\Rightarrow s^2 Y - sf(0) - f'(0) + Y = \frac{1}{s^2}$$

$$\Rightarrow s^2 Y - s \cdot 1 - (-2) + Y = \frac{1}{s^2} \quad \left[\begin{array}{l} Y(0) = f(0) = 1 \\ Y'(0) = f'(0) = -2 \end{array} \right]$$

$$\Rightarrow s^2 y + y - s \cdot 1 + 2 = \frac{1}{s^2}$$

$$\Rightarrow s^2 y + y - s + 2 - \frac{1}{s^2} = 0$$

$$\Rightarrow y(s^2 + 1) - s + 2 - \frac{1}{s^2} = 0$$

$$\Rightarrow y(s^2 + 1) = s - 2 + \frac{1}{s^2}$$

$$\Rightarrow y = \frac{s-2}{s^2+1} + \frac{1}{s^2(s^2+1)}$$

$$= \frac{s}{s^2+1} - \frac{2}{s^2+1} + \frac{1}{s^2} - \frac{1}{(s^2+1)}$$

$$= \frac{s}{s^2+1} + \frac{1}{s^2} - \frac{3}{s^2+1}$$

$$\therefore \mathcal{L}^{-1}\{y\} = y = \frac{s}{s^2+1} + \frac{1}{s^2} - \frac{3}{s^2+1}$$

$$\therefore y = \mathcal{L}^{-1}(y) = \mathcal{L}^{-1} \left[\frac{s}{s^2+1} + \frac{1}{s^2} - \frac{3}{s^2+1} \right]$$

$$= \mathcal{L}^{-1} \left[\frac{s}{s^2+1} \right] + \mathcal{L}^{-1} \left[\frac{1}{s^2} \right] - 3 \mathcal{L}^{-1} \left[\frac{1}{s^2+1} \right]$$

$$\therefore y = \mathcal{L}^{-1}(y) = \cos t + t - 3 \sin t$$

(Ans)

104 Solve the following Initial value problem (IVP) by Laplace Transform $Y'' + 4Y = 12t$, $Y(0) = 0$, $Y'(0) = 7$

Solution: Given that,

$$Y'' + 4Y = 12t$$

Taking the Laplace transform of both sides of the differential equation and using the given conditions,

$$\text{we have, } Y'' + 4Y = 12t$$

$$L\{Y''\} + 4L\{Y\} = 12L\{t\}$$

we have,

$$L\{f''(t)\} = s^2 L\{f(t)\} - sf(0) - f'(0)$$

$$L\{Y''\} + 4L\{Y\} = 12L\{t\}$$

$$s^2 L\{Y\} - sf(0) - f'(0) + 4L\{Y\} = 12L\{t\}$$

$$\Rightarrow s^2 Y - sf(0) - f'(0) + 4Y = 12L\{t\} \quad [\text{Let, } L\{Y\} = Y]$$

$$\Rightarrow s^2 Y - sf(0) - f'(0) + 4Y = 12 \cdot \frac{1}{s^2}$$

$$\Rightarrow s^2 Y - s \cdot 0 - 7 + 4Y = 12 \cdot \frac{1}{s^2}$$

$$\Rightarrow s^2 Y - 7 + 4Y = \frac{12}{s^2}$$

$$\Rightarrow s^2 Y + 4Y - 7 - \frac{12}{s^2} = 0$$

$$\Rightarrow Y(s^2 + 4) = 7 + \frac{12}{s^2}$$

$$\Rightarrow Y = \frac{7}{(s^2 + 4)} + \frac{12}{s^2(s^2 + 4)}$$

$$\Rightarrow Y = \frac{7}{(s^2 + 4)} + 3 \times \frac{4}{s^2(s^2 + 4)}$$

$$= \frac{7}{(s^2 + 4)} + 3 \times \left[\frac{1}{s^2} - \frac{1}{(s^2 + 4)} \right]$$

$$[\text{Let } L\{t\} = \frac{1}{s^2}]$$

$$\left[\begin{array}{l} Y(0) = 0 \\ Y'(0) = 7 \end{array} \right]$$

$$= \frac{7}{(s^2+4)} + \left[\frac{3}{s^2} - \frac{3}{(s^2+4)} \right]$$

$$\therefore L^{-1}(Y) = y = \frac{7}{(s^2+4)} + \left[\frac{3}{s^2} - \frac{3}{(s^2+4)} \right]$$

$$\therefore Y = L^{-1}(Y) = L^{-1} \left[\frac{7}{(s^2+4)} + \left[\frac{3}{s^2} - \frac{3}{(s^2+4)} \right] \right]$$

$$= L^{-1}(Y) = L^{-1} \left[\frac{7}{(s^2+4)} \right] + L^{-1} \left[\frac{3}{s^2} - \frac{3}{(s^2+4)} \right]$$

$$= L^{-1}(Y) = 7L^{-1} \left[\frac{1}{s^2+4} \right] + 3L^{-1} \left[\frac{1}{s^2} - \frac{1}{(s^2+4)} \right]$$

$$= 7L^{-1} \left[\frac{1}{s^2+4} \right] + 3L^{-1} \left[\frac{1}{s^2} \right] - 3L^{-1} \left[\frac{1}{s^2+4} \right]$$

$$= 4L^{-1} \left[\frac{1}{(s^2+4)} \right] + 3L^{-1} \left[\frac{1}{s^2} \right]$$

$$= 4 \cdot \frac{1}{2} L^{-1} \left[\frac{2}{s^2+4} \right] + 3L^{-1} \left[\frac{1}{s^2} \right]$$

$$= 4 \cdot \frac{1}{2} \sin 2t + 3t$$

$$= 2 \sin 2t + 3t$$

$$\therefore Y = L^{-1}(Y) = 2 \sin 2t + 3t$$

(Ans)

105 solve the following initial value problem (IVP)

by Laplace Transform $y'' - 3y' + 2y = 4e^{2t}$,

$$y(0) = -3, \quad y'(0) = 5.$$

Solution:

$$y = f(t)$$

$$\text{Given that, } y'' - 3y' + 2y = 4e^{2t}$$

Taking the Laplace transform of both sides of the differential equation and using the given conditions, we have,

$$y'' - 3y' + 2y = 4e^{2t}$$

$$L\{y''\} - 3L\{y'\} + 2L\{y\} = 4L\{e^{2t}\}$$

we have,

$$L\{f''(t)\} = s^2 L\{f(t)\} - sf(0) - f'(0)$$

And

$$L\{f'(t)\} = s L\{f(t)\} - f(0)$$

$$\therefore L\{y''\} - 3L\{y'\} + 2L\{y\} = 4L\{e^{2t}\}$$

$$\Rightarrow s^2 L\{f(t)\} - sf(0) - f'(0) - 3[sL\{f(t)\} - f(0)] + 2y = 4 \frac{1}{s-2}$$

$$= 4 \frac{1}{s-2} \quad [L\{e^{at}\} = \frac{1}{s-a}]$$

$$\text{and } L\{e^{at}\} = \frac{1}{s-a}$$

$$\Rightarrow s^2 L\{y\} - sf(0) - f'(0) - 3[sL\{y\} - f(0)] + 2y = 4 \frac{1}{s-2}$$

$$\Rightarrow s^2 y - sf(0) - f'(0) - 3[sy - f(0)] + 2y = 4 \frac{1}{s-2} \quad [y = f(t)]$$

$$[L\{e^{at}\} = \frac{1}{s-a}]$$

$$\Rightarrow s^2 y - sf(0) - f'(0) - 3sy - 3f(0) + 2y = 4 \frac{1}{s-2}$$

$$\Rightarrow s^2 y - s\{-3\} - 5 - 3sy + 3(1-2) + 2y = 4 \frac{1}{s-2}$$

$$\Rightarrow s^2 y + 3s - \cancel{3s} y - 5 - 3sy - 9 + 2y = 4 \frac{1}{s-2}$$

$$\Rightarrow s^2 y + 3s - 3sy - 14 + 2y = 4 \frac{1}{s-2}$$

$$\Rightarrow s^2 y - 3sy + 2y + 3s - 14 = 4 \frac{1}{s-2}$$

$$\Rightarrow s^2 y - 3sy + 2y = -3s + 14 + 4 \frac{1}{s-2}$$

$$\Rightarrow y(s^2 - 3s + 2) = -3s + 14 + 4 \frac{1}{s-2}$$

$$\Rightarrow y(s^2 - \cancel{3s} - s + 2) = -3s + 14 + 4 \frac{1}{s-2}$$

$$\Rightarrow y\{s(s-2) - 1(s-2)\} = -3s + 14 + 4 \frac{1}{s-2}$$

$$\Rightarrow y(s-1)(s-2) = -3s + 14 + 4 \frac{1}{s-2}$$

$$\Rightarrow y = -\frac{s}{(s-1)(s-2)} + 14 \frac{1}{(s-1)(s-2)} + 4 \frac{1}{(s-2)} - \frac{1}{(s-1)(s-2)}$$

$$\Rightarrow y = \frac{-3s}{(s-1)(s-2)} + \frac{14}{(s-1)(s-2)} + \frac{4}{(s-1)(s-2)^2}$$

$$\Rightarrow y = \frac{-3s+14}{(s-1)(s-2)} + \frac{4}{(s-1)(s-2)^2}$$

$$= \frac{(-3s+14)(s-2)+4}{(s-1)(s-2)^2}$$

$$= \frac{-3s^2+6s+14s-28+4}{(s-1)(s-2)^2}$$

$$= \frac{-3s^2+6s+14s-24}{(s-1)(s-2)^2}$$

Applying partial fraction,

$$\text{Let, } y = \frac{-3s^2+6s+14s-24}{(s-1)(s-2)^2} = \frac{A}{(s-1)} + \frac{B}{(s-2)} + \frac{C}{(s-2)^2}$$

$$Y = \frac{-3s^2 + 6s + 14s - 24}{(s-1)(s-2)^2} = \frac{-7}{(s-1)} + \frac{4}{(s-2)} + \frac{4}{(s-2)^2}$$

$$\therefore \mathcal{L}^{-1}\{Y\} = \frac{-3s^2 + 6s + 14s - 24}{(s-1)(s-2)^2} = \frac{-7}{(s-1)} + \frac{4}{(s-2)} + \frac{4}{(s-2)^2}$$

$$\therefore Y = \mathcal{L}^{-1}(Y) = -7 \frac{1}{2} \mathcal{L}^{-1} \left[\frac{1}{(s-1)} \right] + 4 \mathcal{L}^{-1} \left[\frac{1}{(s-2)} \right] + 4 \mathcal{L}^{-1} \left[\frac{1}{(s-2)^2} \right]$$

$$\therefore Y = \mathcal{L}^{-1}(Y) = 7e^t + 4e^{2t} + 4te^{2t}$$

(Ans).